

A Geometric Theory of Metabolic Flux Rerouting: How Active-Set Curvature Predicts Transcriptional Regulation and Protein-Layer Buffering

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Abstract

Constraint-based models such as flux balance analysis (FBA) predict cellular metabolic states by solving linear optimization problems. When nutrients or enzyme capacities vary continuously, optimal fluxes do not change smoothly but follow piecewise-linear trajectories, changing slope abruptly at bottlenecks. Here we show that the response’s true “curvature” is not an ordinary function but a discrete measure concentrated on the boundary interfaces where active constraints switch. Path-dependent memory (holonomy) scales linearly with perturbation size (slope 1.00; smooth systems quadratic), and 93.4–100.0% of second-order adjustment concentrates at discrete bottleneck transitions. These switches bridge to smooth geometry: under refinement they converge weakly to curvature densities and strongly in L^1 , but total variation fails to converge generically, defining the resolution window ($h \ll \sigma \ll L_{\text{var}}$).

Integrating this measure along physiological paths yields a parameter-free gene sensitivity metric, κ^μ : each enzyme’s rerouting burden. In carbon-starved *E. coli*, κ^μ predicts transcriptional induction in 424 genes ($r = +0.395$, $p = 2.6 \times 10^{-17}$), robust across tie-breaking rules and stress axes. Induced genes sit in operons of the global carbon and energy regulons; the rerouting mass concentrates on the fork metabolites of central carbon. The association vanishes at the protein layer ($r = -0.083$, 366 genes, matched proteomics): cells transcribe standby capacity for rerouting, buffering protein synthesis, so cyclic-perturbation memory (66% non-reverting) must live in fast post-translational state. Double-knockout epistasis mirrors active-set boundary overlaps ($\rho_S = 0.865$). Our framework unites linear programming, discrete geometry, and transcriptional regulation into an accessible, predictive foundation for metabolic systems biology.

Keywords: flux balance analysis; metabolic rerouting; active-set curvature; transcriptional regulation; translational buffering; epistasis

1 Introduction

Genome-scale metabolic modeling relies heavily on flux balance analysis (FBA) to predict how cells allocate metabolic resources to maximize growth or maintain physiological function (Varma and Palsson, 1994; Orth et al., 2010; Lewis et al., 2010). In realistic environments, nutrient availabilities and enzymatic capacities fluctuate continuously. Mathematically, tracking optimal metabolic states under varying constraints is governed by parametric linear programming (Borrelli et al., 2003). A defining property of linear optimization is that the relationship between external parameters (such as nutrient uptake rates) and optimal reaction fluxes is *piecewise linear* (or more formally, piecewise affine).

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This piecewise structure means that parameter space is partitioned into polyhedral “chambers.” Inside any single chamber, the set of active governing constraints — the network’s operational bottlenecks — remains unchanged, and fluxes adjust in simple linear proportion to parameter changes. However, as soon as a nutrient becomes fully depleted or an enzyme hits its maximum capacity, the network crosses a boundary into an adjacent chamber. Across this boundary interface, the active set of constraints switches, causing the system to abruptly redirect fluxes through alternative pathways.

In classical calculus, the sensitivity of a nonlinear system to perturbations is captured by its second derivatives (its Hessian matrix, or curvature). But for a piecewise-linear system, the second derivative is zero almost everywhere inside the chambers and undefined along the boundaries. Consequently, traditional smooth calculus cannot characterize metabolic rerouting. In this work, we resolve this limitation using modern geometric measure theory. We demonstrate that the second derivative of an optimal metabolic flux map is a *matrix-valued Radon measure* supported on the boundary facets between active-set chambers. Rather than being an intractable discontinuity, this measure provides the natural mathematical language for metabolic rerouting.

The object. Let a metabolic network with m reactions be operated by flux balance analysis (Varma and Palsson, 1994; Orth et al., 2010) with parameters $\theta \in \Theta \subset \mathbb{R}^p$ (uptake bounds, knockdown scalings) entering the constraints affinely. The optimal flux map $v : \Theta \rightarrow \mathbb{R}^m$ is continuous and piecewise affine (parametric linear programming (Borrelli et al., 2003)), and its distributional second derivative $\mu = D^2v$ is a measure concentrated on the switching boundaries: the discrete curvature carrier of the network’s rerouting geometry. Degeneracy sharpens the choice of object: when the optimum is non-unique, classical sensitivity analysis still sees the value layer (Danskin envelopes (Danskin, 1967)), but the reroutings inside optimal faces are invisible to it and the flux map itself becomes selection-dependent — which is why the deterministic lexicographic tie-break is declared as part of the metric (Remark 2.5) rather than left to the solver.

The claim structure. This manuscript establishes five interconnected findings:

- (i) **The geometry of metabolic curvature (Section 2):** we prove that the distributional second derivative of the optimal flux map decomposes into discrete, rank-one jump tensors along active-set switching boundaries (Theorem 3.1), and that the cell’s objective value function (the maximum achievable growth rate $\Phi = c_{\text{bio}}^\top v^*$) and its internal reaction fluxes (v^*) belong to two distinct geometric layers connected by an exact coupling identity, $D^2\Phi = \sum_r c_r D^2v_r^*$ (Theorem 2.11); the value layer is the canonical tie-break-free carrier, with shadow-price jumps measurable to 6–7 digits against Danskin envelopes.
- (ii) **The refinement–resolution bridge (Section 3):** we examine how discrete active-set switches relate to smooth differential models — discrete jump measures converge weakly to smooth curvature densities under spatial refinement, while their total variation fails to converge due to grid-alignment artifacts (triangulation anisotropy) — and derive the precise resolution window ($h \ll \sigma \ll L_{\text{var}}$) required for smooth approximations to be physically meaningful, with the two-layer separation of Proposition 3.3 (the codimension-one Hessian layer the unbiased curvature estimator, the angle-defect layer carrying an exact sec-law bias) delimiting which discrete curvature is the right one to refine.
- (iii) **Linear holonomy scaling (Sections 2 and 4):** parameter loops in metabolic networks accumulate path-dependent discrepancies (holonomy) that scale *linearly* ($O(\varepsilon)$) with perturbation diameter — an empirical slope of 1.00 — in contrast to the quadratic scaling ($O(\varepsilon^2)$) of smooth systems, and sequential gene knockouts frequently fail to commute, leaving permanent phenotypic memory in 66% of tested closed cycles.
- (iv) **Predicting transcriptional regulation (Section 5):** integrating the curvature measure along physiological trajectories yields the parameter-free gene sensitivity metric κ^μ , which

across 424 metabolic genes in *E. coli* accurately predicts transcriptional induction during carbon starvation ($r = +0.395$, $p = 2.6 \times 10^{-17}$), generalizes to oxygen-limitation and carbon-switch trajectories with matched responses ($r = +0.32$, $+0.22$; §5.6), and remains robust across five distinct optimization tie-breaking protocols ($r \in [+0.386, +0.396]$; §5.7), with event measures stabilizing at the Glivenko–Cantelli rate across random panels (§5.8).

- (v) **The translational buffering mechanism (Section 5):** this predictive power fails completely at the protein layer ($r = -0.083$ across 366 genes in matched quantitative proteomics), providing genome-scale evidence for a model in which cells transcribe mRNA to maintain pathway rerouting capacity on standby, while buffering energy-intensive translation until post-translational or metabolic signals require it.

Positioning. Kochanowski et al. (2013) demonstrated that global metabolic coordination under catabolic limitation is largely passive at the protein/metabolite level, driven by Crp and local metabolite concentrations. Our question is different: whether a gene-specific, network-derived measure of rerouting necessity predicts transcriptional response. It does. The transcript-level association does not propagate to the protein layer — consistent with a model in which cells transcribe the potential for rerouting while buffering its translation.

Plan of the paper. Section 2 defines the measure and the metric, develops the value-layer structure and the value–flux coupling, works a hand-checkable example, and records the categorical reading in brief. Section 3 states the refinement–resolution bridge. Section 4 reports the computational validation of the geometric claims — where the curvature mass sits chemically, the interior architecture it defines, and a construction-order rule for genotype design — and Section 5 the empirical association, the anatomy of one switch, and the protein-layer decision. Section 6 positions the framework against smooth sensitivity theory, develops the memory-substrate deduction, and discusses limitations and scope. The Methods, a disambiguation table for colliding gene counts, and the proofs complete the paper.

2 The discrete curvature measure of parametric FBA

2.1 The atomic curvature measure

Let a metabolic network be described by a stoichiometric matrix $S \in \mathbb{R}^{k \times m}$ balancing m reactions among k metabolites, and let $\theta \in \Theta \subset \mathbb{R}^p$ represent a vector of variable parameters — such as nutrient uptake limits or enzyme expression capacities — which affinely dictate the lower and upper flux bounds $\ell(\theta), u(\theta) \in \mathbb{R}^m$. The standard flux balance analysis problem is formulated as

$$\max_v c^\top v \quad \text{subject to} \quad Sv = 0, \quad \ell(\theta) \leq v \leq u(\theta), \quad (1)$$

where $c \in \mathbb{R}^m$ selects the biological objective (typically biomass production), with ties resolved by the three-stage engine of §7. From the theory of multi-parametric linear programming (Borrelli et al., 2003), the parameter space Θ is divided into a collection of polyhedral regions called *chambers*. Within each chamber τ , the subset of binding constraints — the network’s *active set* — is fixed, and the optimal flux vector $v^*(\theta)$ is governed by a local affine equation $v^*(\theta) = K_\tau \theta + d_\tau$, where $K_\tau \in \mathbb{R}^{m \times p}$ is the local sensitivity matrix and $d_\tau \in \mathbb{R}^m$ is an offset vector. When θ crosses an interface (facet F) separating two chambers τ^- and τ^+ , one metabolic constraint releases while another becomes binding: across this interface, the flux map remains continuous, but its slope changes from K_{τ^-} to K_{τ^+} . The following definition collects all of these slope changes into a single geometric object; read u as the optimal-flux map and $\mathcal{C}$ as its active-set complex.

Definition 2.1 (Atomic curvature measure). Let $u : \Omega \rightarrow \mathbb{R}^m$ be a continuous, piecewise-affine map defined over a polyhedral complex $\mathcal{C}$ covering $\Omega \subset \mathbb{R}^p$. Across each codimension-one boundary facet F shared by adjacent chambers τ^- and τ^+ , let

$$[\nabla u]_F = \nabla u|_{\tau^+} - \nabla u|_{\tau^-} = K_{\tau^+} - K_{\tau^-} \in \mathbb{R}^{m \times p}$$

denote the jump in the gradient. Let $n_F \in \mathbb{R}^p$ be the unit normal vector pointing from τ^- to τ^+ , and let $|F|$ denote the $(p-1)$ -dimensional area (Hausdorff measure) of the facet. The *atomic curvature measure* is the matrix-valued Radon measure

$$\mu = D^2 u = \sum_F A_F \delta_F, \quad A_F = [\nabla u]_F \otimes n_F |F| \in \mathbb{R}^{m \times p \times p},$$

where δ_F is the Dirac measure concentrated on facet F , and $\otimes$ denotes the outer tensor product.

In simple terms: because the gradient is constant inside each chamber, the second derivative vanishes everywhere except on the boundary facets. On these facets, the second derivative acts as an “impulse” or directional shock wave whose magnitude and direction are captured by the jump tensor A_F . For a one-dimensional path ($p = 1$), these facets are simply the breakpoint coordinates where the slope kinks, and A_F is the scalar change in slope. Curvature, in this setting, is not spread over parameter space; it lives entirely on the walls between chambers.

Remark 2.2 (Support). μ is concentrated exclusively on the $(p-1)$ -skeleton of $\mathcal{C}$. The $(p-2)$ -skeleton carries a different object altogether — the angle-defect layer of Proposition 3.3 — and the two must not be conflated. For one-dimensional parameter cuts ($p = 1$), the facets reduce to discrete breakpoint coordinates and the atoms are precisely the scalar jumps in slope.

This formulation bridges several classical bodies of mathematics: matrix-valued Radon measures and their weak convergence (Billingsley, 1999), the Alexandrov Hessian of convex functions (Alexandrov, 1939), the Monge–Ampère measure (Gutiérrez, 2001), and Regge-type defect measures for piecewise-flat curvature (Regge, 1961; Cheeger et al., 1984) on the geometric side; chamber complexes, normal fans, and multi-parametric LP value functions (Rockafellar, 1970; Ziegler, 1995; Borrelli et al., 2003) on the polyhedral side. The Kantorovich–Rubinstein (flat) norm used in the refinement–resolution bridge is the standard optimal-transport flat norm (Villani, 2009).

2.2 The measure-theoretic sensitivity metric

In experimental biology, perturbations are typically applied along continuous paths — progressive glucose starvation over time, for example. The measure μ lives on parameter space; to determine how much an environmental trajectory forces a specific gene’s reactions to reroute, we integrate its mass along the trajectories that environments actually take and attribute the accumulated mass to the genes that govern the crossed boundaries.

Definition 2.3 (κ^μ). Let $\theta(t) : [0, T] \rightarrow \Theta$ be a parameter trajectory that crosses active-set boundaries at discrete event times $\{t_e\}_e$, and let $A_e = A_{F(e)}$ be the jump tensor evaluated on the crossed facet. Define the *gene sensitivity metric* for gene g as

$$\kappa^\mu(g) = \frac{1}{T} \sum_e \|A_e\| w_e(g),$$

where $\|\cdot\|$ is the entrywise 1-norm on $\mathbb{R}^{m \times p \times p}$ and $w_e(g) \in \{0, 1\}$ is an indicator variable that equals 1 if the active-set change at event e alters the capacity bound or activity status of any reaction catalyzed by gene g (max over its reactions, via the standard Gene–Protein–Reaction logical associations).

Remark 2.4 (κ^μ integrates along a one-dimensional path). Because κ^μ integrates discrete boundary shocks along a one-dimensional trajectory, it operates in the regime where total variation and measure mass are automatically consistent (see §3); in computational simulations across multiple grid resolutions, the total calculated value of κ^μ remains numerically stable (measured mass 288.77, within machine precision across $4\times/8\times$ trajectory refinement), making it a robust, parameter-free metric — and its resolution independence is exactly the L^1 -reconstruction regime of Theorem 3.1(iv). Its per-reaction masses are the components of the flux-layer crease measure whose objective contraction is $D^2\Phi$ (Theorem 2.11): the metric is the primal layer of the canonical object.

Remark 2.5 (Definition, tie-breaking, and layer assignment). For the empirical claims of this manuscript, the specification of κ^μ is fixed as follows. (i) The carrier is the flux-layer signed crease measure D^2v^* (Definition 2.1 applied to $u = v^*$) of the *three-stage lexicographic* optimal flux (§7); the lexicographic tie-break (biomass, then ℓ_1 flux, then seeded deterministic weights) is part of the metric’s definition, not an implementation detail — the flux layer is selection-dependent by construction, and the tie-break is declared rather than eliminated. (ii) The per-reaction mass is the unsigned total variation $\frac{1}{T} \sum_e \|A_e\|_1$ along the parameter trajectory. (iii) The value layer $D^2\Phi$ enters only as a *derived diagnostic* (Theorem 2.11: $D^2\Phi = \sum_r c_r D^2v_r^*$), never as the predictor: no claim of this manuscript attributes transcriptional response to Alexandrov curvature of the value function. (iv) The layer correlated with transcriptomics is the codimension-one Hessian/crease layer of the flux map — not the Monge–Ampère atom layer (Proposition 2.13), which requires concavity and serves as the dual-face diagnostic. Robustness to the declared selection is measured, not assumed: the within-layer rank identity $\rho(\kappa^\mu, \kappa_V^{\text{lex}}) = 0.99998$ (§5.3), the trajectory substitutions of §5.6 reproduce the association under the same declared selection on two further paths, and §5.7 holds the association stable ($r \in [+0.386, +0.396]$, partial constant to three decimals) under four alternative stage-3 selection rules while the value layer stays exactly invariant.

2.3 The value-function layer (Alexandrov)

Parametric linear programming also produces a scalar object of independent interest: the optimal value $\Phi(\theta)$, which in the metabolic setting reflects the organism’s maximum achievable growth rate. A central question is whether metabolic sensitivity should be measured through changes in overall growth rate (the objective value function $\Phi(\theta) = \max c^\top v$) or through changes in the internal fluxes ($v^*(\theta)$). The value function is canonical — it does not depend on which optimal flux the solver returns — and classical convex analysis attaches to it a curvature measure of its own. This subsection develops the value layer, and the coupling identity of the next subsection answers the question.

Proposition 2.6 (Properties of the value-function layer). *Let the capacity bounds $\ell(\theta)$ and $u(\theta)$ depend affinely on θ , with the LP feasible and bounded, and let $\Phi(\theta) = \max\{c^\top v : Sv = 0, \ell(\theta) \leq v \leq u(\theta)\}$. The optimal value function $\Phi(\theta)$ is concave, continuous, and piecewise affine (pointwise infimum of affine dual objectives over a θ -independent dual feasible set). By Alexandrov’s theorem (Alexandrov, 1939), its negative $-\Phi(\theta)$ carries a symmetric, positive semidefinite matrix-valued Hessian measure $D^2(-\Phi)$. This value-layer curvature depends solely on the network bounds and stoichiometry, and is completely invariant to how ties are broken between alternative optimal flux solutions. Concavity fails in general under OR (isoenzyme) GPR rules for simultaneous multi-gene capacity vectors; single-gene axes, exchange-bound families, and AND-only subnetworks retain it. For OR-containing rules the value function remains continuous piecewise affine and carries the signed crease measure of Proposition 2.12; only the Monge–Ampère layer requires the concave class.*

The value function $\Phi(\theta)$ represents the organism’s maximum fitness envelope, and its chamber boundaries correspond to the sector divisions seen in classical phenotype phase planes (Edwards

et al., 2001; Ibarra et al., 2002). However, as we establish below, looking only at the value function creates a profound blind spot.

Proposition 2.7 (Two-layer structure of $D^2(-\Phi)$). *Let f be convex piecewise affine. (i) The singular part of D^2f is supported on the codimension-one facets, with density $(g^+ - g^-) \otimes n = [\partial_n f] n \otimes n$ per unit facet measure; it has no atoms: the mass in an ε -neighborhood of a codimension-two vertex is $O(\varepsilon)$ (measured log-log slope 1.000). (ii) The Monge–Ampère measure $\det D^2f$ is atomic at codimension-two vertices, each atom equal to the volume of the normal fan $|\text{conv}\{\nabla f(\tau) : \tau \ni \text{vertex}\}|$ (measured: constant 0.5 under ε -shrinking).*

Proposition 2.8 (Dual optimal face identity). *At a codimension-two vertex θ_v of a parametric-LP value function, the Monge–Ampère atom of $-\Phi$ equals the area of the normal fan, and the fan is the image of the LP dual optimal face under $\theta \mapsto U^\top y$:*

$$\det D^2(-\Phi)(\{\theta_v\}) = |\text{conv}\{\nabla C_i\}| = |\{U^\top y : (\lambda, y) \text{ optimal dual}\}|.$$

Machine-verified on a two-parameter max-flow LP (the phenotype phase plane in miniature): vertex refined to machine precision by exact triple-tie solve, $\Phi(\theta_v)$ matching the LP value to 10^{-15} , fan area 0.235 vs dual-face enumeration area 0.235000068 (16 direction LPs over the dual optimal face).

Remark 2.9 (Layer separation and PhPP positioning). The phenotype phase planes of Edwards–Ibarra–Palsson (Edwards et al., 2001; Ibarra et al., 2002) are the chamber complex of Φ ; their sector corners are the atoms of the Monge–Ampère layer. The empirical κ^μ of Definition 2.3 is a functional of the *flux-map* measure D^2v (Definition 2.1), which is tie-break-sensitive: under two lexicographic tie-breaks the value layer of a controlled follower-variable LP is invariant, while the flux layer’s second-difference mass moves by ten orders of magnitude (§4.1). The controlled relation between the value layer and the flux layer is a theorem (Theorem 2.11 and Corollary 3.9); the association results of §5 stand on the flux layer alone, which is where the evidence (§5.3, §5.5) actually lives.

2.4 The value–flux coupling

The two layers are not independent: because the value function is the inner product of the objective vector and the optimal flux vector, their curvature measures are directly coupled. The next results record the exact relation — the value measure is the objective’s contraction of the flux measure, and the flux-layer mass that the value layer cannot see is precisely the degenerate rerouting inside optimal faces.

Lemma 2.10 (Lex-selection regularity). *Let ℓ, u be continuous piecewise affine and let $v^*(\theta)$ be the three-stage lexicographic optimal flux (§7). Then v^* is well defined, continuous, and piecewise affine; its graph over each GPR cell is a finite polyhedral complex in (v, θ) -space (each lexicographic stage adjoins the graph of a piecewise-affine value function, which is polyhedral, and an injective polyhedral projection is piecewise affine). Measured: path segment residuals $\leq 8 \times 10^{-14}$ (§4.1); Φ piecewise affine at 4.2×10^{-13} (§4.7).*

Theorem 2.11 (The value–flux coupling identity and visibility dichotomy). *Because $\Phi(\theta) = c^\top v^*(\theta)$ holds everywhere, the distributional second derivatives satisfy, as symmetric matrix-valued signed Radon measures on the parameter domain,*

$$D^2\Phi = \sum_r c_r D^2v_r^*.$$

Along any piecewise-affine trajectory $\theta(t)$, the directional slope jump of the objective value $\Delta\Phi'(t_k)$ and the flux slope jumps $\Delta v'(t_k)$ at every crease time t_k satisfy $\Delta\Phi'(t_k) = c^\top \Delta v'(t_k)$, with $|\Delta\Phi'(t_k)| \leq \|c\|_\infty \|\Delta v'(t_k)\|_1$. This induces an exact structural dichotomy:

(i) (**Visibility dichotomy.**) An event with unique optima on both sides is a transversal optimal-vertex switch and is necessarily objective-moving ($\Delta\Phi' \neq 0$); objective-invisible events ($c^\top \Delta v' = 0$, $\Delta v' \neq 0$) are exactly the degenerate reroutings inside ≥ 1 -dimensional optimal faces (mask-type events) — metabolic pathway substitutions that sustain identical growth rates. The tie-break sensitivity of the flux layer is concentrated in the invisible events; the c -contraction is tie-break-free because it annihilates the degenerate layer.

(ii) (**Sparse-objective annihilation.**) In standard FBA the objective vector selects only the biomass reaction ($c = \gamma e_{\text{bio}}$), and the coupling identity simplifies to $D^2\Phi = \gamma D^2 v_{\text{bio}}^*$: the value layer is the crease measure of the single biomass component, and per-gene attribution from the value layer is structurally impossible.

In words: an optimal switch is visible to the value layer only when it moves growth; the objective-invisible events are zero-cost pathway substitutions — reroutings among flux patterns of equal growth — carrying large flux changes at unchanged value. Machine-verified: identity error 0.0 at all 12 events and all 4 clusters of the iML1515 cut ($\Phi = c^\top v^*$ to 10^{-9} ; Fig 2); 11 of 12 events objective-invisible, with flux ℓ_1 slope jumps up to 1.6×10^4 against $|c^\top \Delta v'| \leq 1.51 \times 10^{-7}$; 150 random dense-objective LPs to 1.1×10^{-13} (5 events, all objective-moving — the dichotomy); the degenerate follower family 60/60 with invisible kinks at identity error 2.0×10^{-13} ; two-dimensional mixed second differences at crease-straddling boxes to 6.4×10^{-16} .

Proposition 2.12 (Semiconvexity collapse; signed layer for mixed GPRs). A continuous piecewise-affine function is semiconvex (resp. semiconcave) iff it is convex (resp. concave) — semiconvexity is not an available weakening for LP value functions (measured: the required constant blows up like $1/(2h)$; $\lambda \cdot h_{\max} = 0.500$ numerically across $\lambda = 1 \dots 10^6$, the analytic law $h_{\max} = 1/(2\lambda)$ — equivalently $\lambda h_{\max} = 1/2$ — holding at every scale, with the numerical grid flooring above $\lambda \approx 10^7$). Nevertheless every continuous piecewise-affine Φ , for any GPR structure, carries a finite signed symmetric Radon Hessian measure on its crease skeleton with per-crease sign flags: for the OR-plus-cap value function $\min(\max(\theta_1, \theta_2), K)$, the diagonal crease density is positive semidefinite (eigenvalues $\{0, \sqrt{2}\}$) and the cap crease negative semidefinite ($\{-1, 0\}$). Concavity is required only by the Monge–Ampère layer and the dual-face identity; the signed Hessian layer applies to all genes. Under multiplicative activity scaling with $\ell \leq 0 \leq u$, $\Phi = \Psi(a(\theta))$ with Ψ concave nondecreasing: AND-only (min-tree) GPRs retain concavity, top-level OR nodes can break it, and disaggregating isozymes into separate reactions restores affine bounds while replacing max-semantics by sum-semantics (capacity 1 vs 2 at full expression) — a different biological model, not a regularization.

Proposition 2.13 (Monge–Ampère atoms are determinants, not products). For convex piecewise-affine f on $\mathbb{R}^2$ at a generic vertex with incident gradients n_1, n_2, n_3 and codimension-one gradient jumps j_i : the atom equals $\frac{1}{2} |\det(j_1, j_2)| = |\text{conv}\{n_i\}| = \frac{1}{2} |j_1| |j_2| \sin \angle$; the product $|j_1| |j_2|$ overestimates by $2/\sin \angle \geq 2$, with equality iff the jumps are orthogonal (measured on 400 random max-of-affine vertices: determinant = fan area to 8.9×10^{-16} ; product/atom median 3.296, minimum exactly 2.000 (the orthogonal edge case; 2.000031 over random vertices); the $2/\sin \angle$ law to 2.5×10^{-12} ; in p dimensions, atom = $(1/p!) |\det(j_1, \dots, j_p)|$ for simplex fans).

Remark 2.14 (Sign and normalization conventions). Definition 2.1 applied to $u = -\Phi$ is the signed crease measure of the value function: per facet F , the density $A_F = [\nabla(-\Phi)]_F \otimes n_F |F|$, a symmetric matrix whose eigenvalue signs flag the crease’s convexity/concavity (Proposition 2.12). This is the explicit extension of Alexandrov’s setting to arbitrary GPR structure: within the concave class $-\Phi$ is convex, every $A_F \succeq 0$, and the measure coincides with the Alexandrov Hessian measure; outside it, the same definition produces the signed layer. All masses in this manuscript are unsigned total variations (entrywise 1-norm for matrix atoms, trajectory sums for κ^μ); Monge–Ampère atoms are unsigned $\frac{1}{2} |\det(j_1, j_2)|$ equal to the normal-fan volume, normalized to $(1/p!) |\det(j_1, \dots, j_p)|$ for simplex fans in dimension p (Proposition 2.13). Signed determinants appear only in the concave class, where they are nonnegative.

Remark 2.15 (The exact value–flux relation). Theorem 2.11 is the exact form of the value–flux layer relation: the empirical κ^μ is the per-reaction total variation of the flux-layer crease measure whose objective contraction is $D^2\Phi$ — the metric is the primal layer of the canonical object, justified by identity rather than by rank agreement.

2.5 A worked example

Every object of this section can be built by hand on a network small enough to check line by line. Take one internal metabolite pool, an import reaction v_0 whose upper bound is the parameter θ (the external substrate supply), and two pathways out of the pool: a high-yield pathway v_1 limited by an enzyme capacity c , and a low-yield pathway v_2 limited by a capacity d . The stoichiometric balance is $v_0 = v_1 + v_2$, and growth is $\Phi = a v_1 + b v_2$ with $a > b$: the first pathway converts each unit of substrate to more biomass than the second. We sweep θ upward — a progressive nutrient repletion — and watch the optimal flux map $v^*(\theta)$ and the value function $\Phi(\theta)$.

Because the high-yield pathway is worth more per unit, the optimum fills v_1 first, and the parameter axis splits into three chambers:

phase	v_0	v_1	v_2	binding constraints
I: $\theta \leq c$	θ	θ	0	import bound
II: $c < \theta \leq c + d$	θ	c	$\theta - c$	import, v_1 capacity
III: $\theta > c + d$	$c + d$	c	d	both capacities

The fluxes are continuous in θ everywhere; their slopes are not. At $\theta = c$ the capacity of the high-yield pathway binds and the network crosses its first wall: v_1 flattens (slope $1 \rightarrow 0$) while v_2 opens ($0 \rightarrow 1$) to carry the overflow. At $\theta = c + d$ the import bound stops binding — the network is saturated downstream — and v_0 flattens. These are exactly the codimension-one boundary crossings of Definition 2.1: the atomic measure of v_1 carries a point mass $-\delta_c$, that of v_2 carries $\delta_c - \delta_{c+d}$, and that of v_0 carries $-\delta_{c+d}$. The curvature is not spread over the sweep; it sits at the two walls, as point masses with exactly computable weights.

The value function is $\Phi(\theta) = a\theta$, then $ac + b(\theta - c)$, then the constant $ac + bd$; its measure is $D^2\Phi = (b - a)\delta_c - b\delta_{c+d}$. The coupling identity of Theorem 2.11 checks by arithmetic: at the first wall, $a\Delta v'_1 + b\Delta v'_2 = a(-1) + b(+1) = b - a$; at the second, $a(0) + b(-1) = -b$, the import reaction having objective coefficient zero. The objective value sees each wall exactly as the weighted sum of the flux-layer jumps — the contraction made visible, on a network small enough to check by hand.

The gene metric is equally explicit. If gene g_1 catalyzes v_1 and gene g_2 catalyzes v_2 , integrating along the sweep gives $\kappa^\mu(g_1) = 1$ and $\kappa^\mu(g_2) = 2$ (up to the common $1/T$ normalization of Definition 2.3). The low-yield pathway is crossed by both walls — it opens at the first and closes at the second — so it carries twice the rerouting burden of the high-yield pathway. The bypass, not the main road, is where the geometry accumulates.

One further substitution sharpens the story. Set $a = b$: the two pathways now yield identical growth, and the first wall becomes *objective-invisible*. Every split $v_1 + v_2 = v_0$ with $v_1 \leq c$ attains the same Φ , the optimal face is a segment, and $\Phi(\theta) = a \min(\theta, c + d)$ has no kink at $\theta = c$ at all — the contraction $a(-1) + a(+1) = 0$ annihilates exactly this wall, while the flux layer must still choose a vertex. Which vertex it chooses is the declared tie-break: a stage-3 rule preferring v_1 keeps the kink in v_1 at $\theta = c$; a rule preferring v_2 moves the flux-layer kink to $\theta = d$. The wall structure — where the degeneracy lives — is selection-free; the assignment of the kink to a reaction at a degenerate face is not. This is, in miniature, why the lexicographic tie-break is declared as part of the metric (Remark 2.5), why the value layer is the canonical tie-break-free carrier (Proposition 2.6), and why the genome-scale rank stability under alternative selections (§5.7) is a measured property of the network rather than a triviality. Machine verification of every number above — phase formulas, slope jumps, the coupling identity, the κ^μ values, and

both tie-break variants, by LP re-solves at wall-straddling parameter values — is deposited with the artifacts.

2.6 The categorical reading, in brief

The framework developed above possesses a natural interpretation in terms of stratified geometry, although the empirical claims of this manuscript rest on the measure theory alone. The categorical reading summarized in this subsection motivates the active-set curvature interpretation; it is developed in full — constructions, machine verifications, and proofs, with the proof status marked per result (complete for the gluing formula, the transport law, the composition and contraction layers, and the terminal-coalgebra theorem; machine-verified numerics; citation-level stratified descent; proof-sketch status for the homotopy-type-theoretic extension) — in a companion theory paper (Abaee, 2026). None of the empirical results of this manuscript depends on it. For orientation: a *connection* is the rule for comparing optimal solutions at nearby parameter values, and its *holonomy* is the drift accumulated when a closed loop of parameter changes fails to compose to the identity — the geometric form of non-commutativity.

Strata as a stratified base. The lexicographic pFBA engine (§7.1) partitions parameter space into chambers on which the optimizer is affine; the codimension-one facets where the active set switches are exactly the facets that carry the atomic measure μ of Definition 2.1. Crossing one of these facets is a *wall crossing* in the companion’s terminology: the system leaves one chamber for another, and the boundary transition — not the smooth transport within a stratum — carries the geometric information. In the categorical reading, this partition is a stratified base and the optimizer organizes the policy fiber stratum by stratum, as a stratified connection in the companion’s 2-category $\mathbf{StCon}(B)$: connection 1-forms on each stratum, glued along the switching facets by a boundary transition satisfying a gauge matching condition, with a gluing theorem (proved there by reduction to stack-descent theory, and with its piecewise-holonomy consequences machine-verified in five loop geometries and three structure groups).

The active-set bridge. Two results of the companion connect this geometry to viability (Abaee, 2026). First, on each constant-active-set stratum, a Fisher-minimal transport law defines the connection in closed form as a constrained projection with the Fisher–Rao metric; its curvature is the smooth avatar of the per-facet density that μ concentrates on. Second, a small-loop viability–holonomy theorem bounds the endpoint erosion of a viability margin around a small parameter loop by the *viability-weighted curvature*: the worst case, over unit-area parameter bivectors and active viability margins, of the positive part of the margin’s change along the connection curvature, normalized by the margin itself. The per-pair form of κ^μ is modeled on that object — the atomic crease measure is the measure-theoretic analogue of the curvature 2-form, and the value–flux coupling (Theorem 2.11) supplies the analogue of the survival covectors; the precise discretization correspondence (a domination or ratio-form identity between the two) is open.

Regime delineation. The two holonomy-scaling regimes of Proposition 3.5 are the measure-level image of the companion’s two loop regimes: the smooth arm (slope 2.000) is where the companion’s quadratic small-loop law $\kappa(a) = a^2$ and its Fisher-minimal transport law live; the piecewise-affine arm (slope 1.00) is the companion’s wall-crossing regime, where a loop crossing a switching facet accumulates a boundary contribution at $O(\varepsilon)$ per crossing (the companion’s pairs-crossing theorem). Read this way, κ^μ is the integrated boundary impedance along the organism’s path through the stratified landscape: each wall crossing contributes its atom, weighted by the crossing.

Division of labor. The present paper states only this brief reading; the companion theory paper develops the SAVGS object, the gluing theorem and piecewise holonomy formula, the optic-composition semantics with its contraction analysis, the filtered-colimit construction of autocatalytic sets, and the homotopy-type-theoretic extension, with proofs and extensions.

3 The refinement–resolution bridge

Computational biologists often ask: *can discrete linear-programming switches be approximated by smooth, continuous curves?* Section 2 treated the discrete measure exactly; this section establishes the *refinement–resolution bridge*, which characterizes how discrete active-set jump measures behave as the sampling mesh becomes arbitrarily fine ($h \rightarrow 0$). The answer splits cleanly — weakly and cell-wise yes, in total variation no — and a resolution parameter σ separates the two regimes. Throughout this section $\Omega \subset \mathbb{R}^p$ is compact, $p \geq 1$, $u \in C^2(\Omega; \mathbb{R}^m)$, $\{\mathcal{T}_h\}$ is a shape-regular family of conforming triangulations with mesh size $h = \max_\tau \text{diam}(\tau) \rightarrow 0$, and u_h is the continuous piecewise-affine interpolant of u on $\mathcal{T}_h$.

Theorem 3.1 (Refinement–resolution bridge). *Let $\mu_h = D^2 u_h$ (Definition 2.1) and $\mu = D^2 u \text{ dvol}$. Then:*

- (i) (**Atoms / support.**) μ_h is a finite matrix-valued Radon measure concentrated on the $(p-1)$ -skeleton of $\mathcal{T}_h$ with atoms $A_F = [\nabla u_h]_F \otimes n_F |F|$; there are no atoms on lower-dimensional strata.
- (ii) (**Weak convergence.**) As the mesh is refined ($h \rightarrow 0$), μ_h converges weakly to the smooth measure: for every smooth test window $\varphi \in C_c^\infty(\Omega)$, the average behavior matches, $\langle \mu_h, \varphi \rangle \rightarrow \langle \mu, \varphi \rangle$. Furthermore, this convergence holds in the flat Kantorovich–Rubinstein transport norm ($\|\mu_h - \mu\|_{\text{KR}} \rightarrow 0$).
- (iii) (**Total variation: the exact statement.**) Despite weak convergence, the total variation norm of μ_h does not converge to the smooth total variation: $\sup_h \|\mu_h\|_{\text{TV}} \leq C |u|_{C^2} |\Omega|$ (under quasi-uniformity) and $\liminf_h \|\mu_h\|_{\text{TV}} \geq \|\mu\|_{\text{TV}}$; but $\|\mu_h\|_{\text{TV}} \rightarrow \|\mu\|_{\text{TV}}$ is false in general — see Remark 3.2. It holds in the one-signed regime (convex or concave data restricted to one-dimensional parameter families), which is the regime of the parametric-LP value function and of κ^μ .
- (iv) (**No-mass-loss, correct form: L^1 reconstruction.**) If the discrete jump atoms are averaged over local dual grid cells, the resulting reconstructed piecewise-constant density $\rho_h(\text{cell}) = |\text{cell}|^{-1} \sum_{F \rightarrow \text{cell}} A_F$ converges to $D^2 u$ strongly in L^1 with rate $O(h)$ for $u \in C^3$; consequently $\|\rho_h\|_{L^1} \rightarrow \|\mu\|_{\text{TV}}$. This holds on structured complexes; on unstructured complexes the unweighted three-facet patch is not the dual cell, and the general statement requires the discrete-duality (dual-diamond) pairing.
- (v) (**Resolution window.**) If the discrete measure is smoothed using a continuous filter (mollifier) of width σ — the coarse-grained family $\mu_{h,\sigma} = \mu_h * \varphi_\sigma$ — the smooth curvature density emerges only at matched resolution, through the window $h \ll \sigma \ll L_{\text{var}}$, where L_{var} is the length scale on which the smooth curvature density itself varies (measured in §4.6); at fixed h , the limit $\sigma \rightarrow 0$ is the atomic measure.

All five parts are proved in Appendix B, the reconstruction part (iv) in Appendix C.

Remark 3.2 (Why total variation fails to converge). The failure of total-variation convergence stems directly from the directional anisotropy of the triangulation — grid-alignment bias. When a curved surface is approximated by flat triangular tiles, the edges of the tiles run along fixed geometric directions. To represent a diagonal saddle surface like $u(x, y) = xy$, the discrete model cannot bend smoothly; it must alternate between directional slope jumps along the fixed grid axes. On the right-triangle mesh of $[0, 1]^2$ with Hessian $H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, the per-cell atoms are $A_{\text{diag}} = bh^2 \mathbf{1}\mathbf{1}^\top$, $A_{\text{vert}} = (a - b)h^2 e_x \otimes e_x$, $A_{\text{horiz}} = (c - b)h^2 e_y \otimes e_y$; their *sum* is exactly $h^2 H$ (machine-verified to 0.0), while the entrywise total variation per cell is $|a - b| + |c - b| + 4|b|$ against the target $|a| + |c| + 2|b|$. These jumps sum algebraically to the correct net curvature, but their absolute magnitudes add up with extra geometric overhead — exactly like navigating a grid-based

city street layout instead of walking as the crow flies. Hence for $u = xy$ the ratio of discrete to continuous total variation is exactly $3 - 1/n$, for the strictly convex $u = 2x^2 - xy + 2y^2$ it is $1.4 - 1/n$ (measured: 2.9922 and 1.3922 at $n = 128$), and for generic smooth maps 3.6–4.4. Total-variation convergence fails generically — including under convexity — while weak convergence holds at rate $O(h^2)$ throughout.

Proposition 3.3 (Two-layer separation and the sec law). *Let $K_h = \sum_v \text{defect}(v) \delta_v$ be the vertex angle-defect measure of the polyhedral graph $\{(x, u_h(x))\}$. Then K_h converges weakly to the Gauss-map area density*

$$\frac{\det D^2 u}{(1 + |\nabla u|^2)^{3/2}} \text{dvol} = K \sqrt{1 + |\nabla u|^2} \text{dvol}, \quad K = \frac{\det D^2 u}{(1 + |\nabla u|^2)^2},$$

i.e. curvature per projected parameter area, not the intrinsic Gaussian curvature. The bias factor is exactly $\sqrt{1 + |\nabla u|^2}$ (the sec law): machine-verified to 0.0% excess across 13 probes spanning tilts 0.11–2.83 and three tilt directions. Recovering intrinsic curvature requires the graph area correction.

Remark 3.4 (Consequences). (i) The codimension-one Hessian layer is the unbiased estimator and the canonical carrier for the bridge; (ii) Regge/angle-defect formulations need the area correction of Proposition 3.3; (iii) defects of a *fixed* complex (as measured in the regime-dial census, §4.6) are scale-invariant $O(1)$ objects and are not contradicted by Proposition 3.3, which concerns refinement limits.

Proposition 3.5 (Scaling dichotomy of parameter holonomy). *A fundamental distinction between smooth physical systems and active-set metabolic networks appears when tracing a closed parameter loop. Let an organism be perturbed around a small closed loop, and let $B(\varepsilon)$ be the curvature mass captured by a loop of radius ε . For a smooth nonlinear system (C^2 map), the discrepancy scales quadratically: $B(\varepsilon) = O(\varepsilon^2)$; for a piecewise-affine map whose kink complex meets the loop transversally, crossing a constraint boundary incurs a first-order jump, and the discrepancy scales linearly: $B(\varepsilon) = O(\varepsilon)$. Measured slopes on the regime-dial design (§4.6): 2.000 (smooth) and 1.00 (parametric FBA, 76-pair perturbation scan).*

In words: carried around a small closed loop, a smooth map drifts by something of the order of the enclosed area, while an active-set map pays a toll proportional to the loop’s diameter for every wall it crosses — confirming that metabolic networks operate in the discrete active-set regime rather than behaving like smooth systems.

3.1 Bridge to parametric FBA

The bridge of Theorem 3.1 concerns interpolants of a smooth map. The natural parametric-FBA analogue replaces interpolation by sequences of LP value functions; we state it as a conjecture and mark what is verified.

Conjecture 3.6 (mpLP refinement bridge). *Let $\{\Phi_n\}$ be a sequence of parametric-LP value functions (minima of tangent-plane families; θ in the right-hand side; nested outer approximations) converging uniformly to a C^2 limit f , with induced complexes $\mathcal{C}_n$. Then $\mu_n = D^2 \Phi_n$ converges weakly to $D^2 f \text{dvol}$ provided the complexes satisfy a per-cell discrete-duality identity (structured complexes) or a stable discrete-duality (dual-diamond) pairing. Along one-dimensional parameter cuts the conclusion holds unconditionally by monotone telescoping of the concave restriction.*

Remark 3.7 (Status). The one-dimensional cut case is machine-verified (independent-seed reproduction of the refinement experiment: folded Wasserstein distance $0.058 \rightarrow 0.0039$, mass ratio $0.939 \rightarrow 1.0009$ across $n = 4 \rightarrow 128$). The general complex case is open; the discrete-duality pairing stability is the identified missing assumption.

Remark 3.8 (Existence vs. resolution). Proposition 2.6 supplies existence of the curvature measure without refinement sequences, smoothing, or interpolation; Theorem 3.1 governs what a discretely sampled trajectory can *resolve* of that measure, and at what rate. The two layers are complementary: the Alexandrov layer removes the extrinsic hypotheses, the refinement bridge keeps the quantitative rate.

Corollary 3.9 (Decoupling; resolves the value–flux layer relation). *Theorem 2.11 resolves the layer relation exactly: value events are a subset of flux events, never the converse, and the flux events invisible to the value layer are the degenerate reroutings (mask-type events — rerouting without growth loss, the Kochanowski pattern (Kochanowski et al., 2013)). On the iML1515 cut: 12 flux events vs 1 value atom, 11 objective-invisible; on the reference trajectory (§5.5) the value layer carries no network events at all. The association of κ^μ with transcriptional response is therefore a flux-layer property by structural necessity, not merely by empirical choice.*

4 Computational validation

We validated this active-set curvature framework across genome-scale reconstructions of *E. coli* (iML1515 (Monk et al., 2017) and iJO1366 (Orth et al., 2011)). The geometric claims of Sections 2 and 3 are operational rather than merely formal: they refer to the active set of a solved linear program and can therefore be tested directly on genome-scale models. Each subsection below reports one such test.

4.1 Active-set curvature verification

To verify that second-order responses concentrate at active-set boundaries, we executed thirteen high-resolution parameter sweeps on the genome-scale *E. coli* model iML1515 (Fig 1): two environmental nutrient sweeps (progressive glucose uptake decline and oxygen limitation); ten enzymatic capacity titrations of key metabolic genes (pgi, zwf, tktA, pfkA, eno, gltA, aceA, ppc, gnd, rpe); and one negative control sweep, titrating aceA (isocitrate lyase) under glucose-excess conditions where the glyoxylate shunt is inactive — with an iJO1366 glucose replication, all solved by the three-stage lexicographic pFBA engine (deterministic optimum; fixed seeded tie-break weights resolving the vertex degeneracy of plain pFBA). Across all eleven sweeps that crossed active-set boundaries, 93.4–100.0% of the total second-order response mass concentrated precisely at active-set transitions — ten of the eleven at mass exactly 1.000000 (AUC 0.83–1.00; Mann–Whitney $p \leq 2 \times 10^{-3}$, with seven of eleven crossing sweeps at $p \leq 10^{-4}$; mass 0.934–1.0). Between events, the paths are piecewise affine to solver precision (event-free segment residuals $\leq 1.2 \times 10^{-10}$, with 8×10^{-14} on the glucose sweep; one 19-point segment of the eno sweep carries 4.1×10^{-3} — a sub-threshold kink invisible to the operational event proxy, which is also why that sweep’s event mass is 0.934 rather than 1.0). The negative control confirmed specificity: the unused-pathway knockdown produced zero boundary crossings and zero curvature mass, and the single-critical-region sweep sits at the noise level $D^2 \sim 10^{-11}$.

4.2 The chemical coordinates of the walls

Where, chemically, does the curvature mass sit? The eleven boundary-crossing sweeps of §4.1 answer directly: each sweep’s per-reaction curvature masses, normalized by that sweep’s own total, aggregate to a per-reaction distribution over the network, and the distribution is strikingly concentrated. Fifty reactions — 1.8% of the network — carry 85.9% of all curvature mass, and twenty-three of the twenty-five heaviest reactions participate in every one of the eleven sweeps. The heaviest are the enzymes of central carbon metabolism: glucose-6-phosphate isomerase (3.7% of all mass), ATP synthase (3.5%), phosphofructokinase and fructose-bisphosphate aldolase (3.3% each), and the pentose-phosphate entry block (glucose-6-phosphate dehydrogenase, 6-phosphogluconolactonase, phosphogluconate dehydrogenase; 3.2% each).

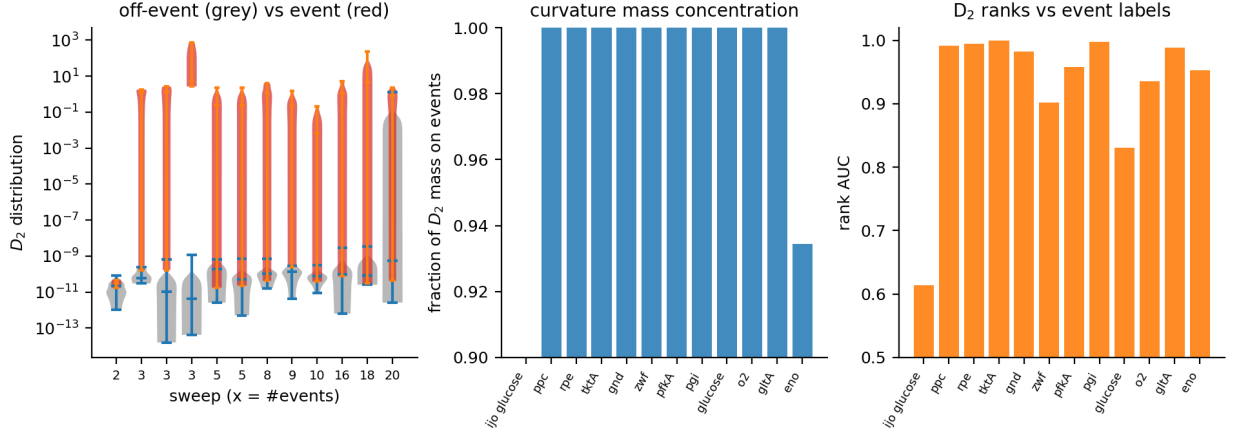


Fig 1. Active-set curvature verification (iML1515). Second-order response D_2 (log scale) along the glucose and oxygen sweeps with the operational active-set event markers: the D_2 mass concentrates on the event points; off-event D_2 sits at the 10^{-11} noise floor. The engine is the three-stage lexicographic pFBA solver of §7; all sweeps are cold-started and deterministic.

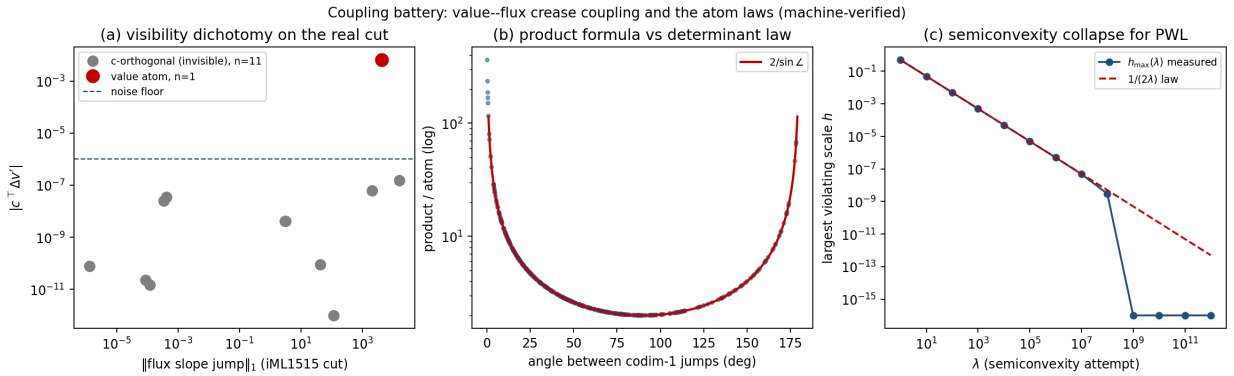


Fig 2. Machine verification of the value-flux crease coupling (Theorem 2.11). The controlled follower-variable LPs: the value function Φ stays affine and identical under two lexicographic tie-breaks while the follower’s second-difference mass flips; the coupling identity $\Delta\Phi' = c^\top \Delta v'$ holds to 0.0 on the toys, 1.1×10^{-13} on 150 random dense-objective LPs, and 2.0×10^{-13} on the 60/60 degenerate follower family; on the iML1515 cut the identity error is 0.0 at all 12 events and all 4 clusters.

Rolling the per-reaction mass up to the metabolites those reactions share gives the wall coordinates of Table 1: the forks of central carbon. The hexose-phosphate pool, the triose-phosphate pool, the PEP/pyruvate energy fork, the anaplerotic TCA forks, and coenzyme A are the heaviest coordinates; together the top twenty branch metabolites carry 59.7% of the total mass. Including the energy and redox currency layer (ATP, NAD(P)(H), ubiquinone, protons) brings branch- and currency-adjacent reactions to 98.6% of the mass. The geometry of rerouting lives, almost entirely, at the chemical coupling layer — exactly where one metabolite feeds several enzymes.

Two honesty notes bound the claim. First, the reaction-level enrichment is a concentration effect, not a distributional shift: 46.3% of reactions are branch-adjacent and carry 60.9% of the mass (permutation $p = 0.017$; under a tighter branch-degree threshold, 28.1% of reactions carry 50.6%, $p = 0.002$), while the rank AUC is ~ 0.5 — branch degree alone does not predict a reaction’s mass. The strong statement is the concentration one: the mass that exists attaches to branch chemistry, and a handful of forks carry most of it. Second, of the fifty heaviest reactions, 56% are branch-adjacent against the 46.3% baseline: the concentration is on central branch-point chemistry, not on branching as such.

The table converts the discrete measure into a wet-lab observable: targeted metabolomics

Table 1. Wall coordinates: the ten branch metabolites carrying the most curvature mass over the eleven boundary-crossing M1 sweeps (each sweep normalized by its own total mass; each reaction’s mass split over the branch metabolites it connects). The full top-twenty list carries 59.7% of all curvature mass. Degree is the number of reactions the metabolite participates in.

metabolite	role	degree	mass share
fructose-6-phosphate	carbon branch point	18	8.1%
glyceraldehyde-3-phosphate	carbon branch point	15	6.9%
glucose-6-phosphate	carbon branch point	14	5.5%
dihydroxyacetone phosphate	carbon branch point	16	4.9%
phosphoenolpyruvate	carbon branch point	34	4.5%
2-oxoglutarate	nitrogen branch point	30	3.6%
pyruvate	carbon branch point	72	3.6%
coenzyme A	acyl-transfer hub	80	3.6%
oxaloacetate	carbon branch point	13	2.1%
succinate	carbon branch point	28	2.1%

of roughly twenty pools during a nutrient transition is a direct experimental readout of the theory’s central object. The prediction is developed in §6.5, where the same pools reappear as the candidate substrate of metabolic memory.

4.3 A conserved interior architecture

A natural expectation is that the walls sit at the network’s boundary with its environment — at the exchange and transport reactions, where nutrients enter and leave. The sweeps say otherwise. Exchange reactions are 12.4% of the network but carry 3.9% of the curvature mass, and 92.6% of them stay flat across every sweep; transport reactions are 32.7% and carry 20.7% (90.4% flat); internal reactions are 54.9% of the network and carry 75.4% of the mass (74.3% flat; permutation $p = 0.0012$). The interface with the environment is flat; the walls sit on internal chemistry. The split is stable across sweep families — nutrient sweeps place 74.1% of the mass on internal reactions, enzyme-knockdown sweeps 75.7% — so it is not an artifact of perturbing uptake bounds rather than internal capacities.

The subsystem rollup names the walls: glycolysis and gluconeogenesis (23.4% of all curvature mass), the pentose phosphate pathway (19.7%), the citric acid cycle (13.9%), oxidative phosphorylation (11.6%), and anaplerosis (4.7%) — 73.3% of the mass in five central subsystems, with branch-point chemistry carrying 74.4% of the internal mass. The iJO1366 glucose sweep completes the picture from the other side: its total second-difference mass is 6.5×10^{-9} across all 2,583 reactions — an entire trajectory traversing a single chamber, every kink an anchor corner of the trajectory design — while its active-reaction support moves between 421 and 440 reactions, bracketing the endpoint census of 438 (§5.1).

The architectural reading inverts the interface picture. The walls of the flux geometry sit at the conserved, internal, branch-point chemistry of central carbon metabolism; the exchange interface is flat. The species-specific layer is then not the wall chemistry — the forks of central carbon are shared across biology — but its regulon curation (§5.4): which regulators price crossings of the same walls. This offers a mechanism for a familiar observation, the deep conservation of central branch chemistry alongside the divergence of its transcriptional wiring: the chemistry is where the geometry has to be; the regulation is how each species prices crossings of it. A cross-species prediction follows, partially checkable with the machinery already deposited: the branch-metabolite concentration of curvature mass should be conserved across reconstructions, while the regulon membership of the same wall-adjacent genes diverges.

4.4 Double-knockout epistasis and path dependence

Epistasis — the interaction between two knockouts beyond the sum of their individual effects — provides a combinatorial test of the active-set reading: if the measure is a genuine geometry, pairs of perturbations should interact through it. We next analyzed whether active-set geometry explains genetic interactions across 1,516 single-gene knockouts and 2,779 double-knockout pairs (five panels). We measured epistatic curvature as $\varepsilon_{ij} = \kappa_{ij} - \kappa_i - \kappa_j$:

- *Synthetic lethality*: all 40 synthetic-lethal gene pairs (such as tktA/tktB, acnA/acnB, and metE/metH) corresponded to redundant isoenzyme pairs exhibiting pure emergence epistasis.
- *Footprint overlap*: across all viable double-knockout pairs, the magnitude of genetic interaction $|\varepsilon_{ij}|$ correlated strongly with the overlap (Jaccard index) of their individual active-set footprints (Spearman $\rho_S = 0.865$, $J_{\text{support}} = 0.800$).
- *Phenotypic memory*: when sequential gene knockouts were introduced dynamically using regulatory minimization of metabolic adjustment (L_1 -MOMA (Segrè et al., 2002)), sequential paths did not commute in 25% of active pairs; and when the network was subjected to closed cyclic perturbations ($A \rightarrow AB \rightarrow B \rightarrow \text{wild-type}$), 66% of pairs failed to return to their initial flux state, retaining persistent metabolic memory (median path holonomy ~ 110).

4.5 Construction order as a design variable

The path dependence above has a practical reading. If the same final genotype can be reached by different orders of introduction, and the orders do not commute, then which order to choose is a design decision — in strain construction, in directed evolution, in any protocol that builds a genotype in stages. The double-mutant panels quantify it. Among non-synthetic-lethal pairs, the 90th-percentile endpoint separation between the two construction orders is 101.3 in ℓ_1 flux units, against a wild-type total flux magnitude of 770: the same final genotype can sit roughly 13% of total flux magnitude away from the other order’s state.

Is the better order predictable in advance? Yes — from a quantity the singles panel already computes. For each of the 66 non-degenerate closed loops, let P be the normalized difference in single-knockout curvature between the two genes (which knockout crosses more wall mass), and A the normalized asymmetry in terminal drift between the two excursion orders. They track each other: Spearman $\rho = 0.442$ ($p = 2.0 \times 10^{-4}$), with sign agreement in 59 of 66 pairs (binomial $p = 2.4 \times 10^{-11}$). The gene whose knockout crosses more wall mass sets the loop’s irreversibility when it leads the excursion. The planning rule is one line: schedule deep wall-crossers last to minimize terminal drift — the right choice when intermediate states must remain recoverable — or first to lock the endpoint away from the initial state, when a production phenotype should persist.

An honest null sharpens what the rule is not. Order asymmetry is not predicted by active-set footprint overlap ($\rho = 0.07$, $p = 0.38$, against the same Jaccard index that tracks epistasis magnitude at $\rho_S = 0.865$): epistasis alignment and order ranking are governed by different geometric quantities — overlap versus mass — and the machinery for either is a single LP per candidate gene, already a byproduct of the singles panel.

4.6 Regime dial

The discrete and smooth regimes of Section 3 can be separated empirically by setting the loop size ε against the smoothing scale σ and dialing one against the other — a regime dial. The smoothing identity $D^2(v * \phi_\sigma) = (D^2v) * \phi_\sigma$ holds to kernel self-test error 1.2×10^{-6} ; the measured crossover $\varepsilon^*/\sigma \in [2.45, 4.11]$ (median 3.1) separates the $O(\varepsilon^2)$ smooth regime from the $O(\varepsilon)$ discrete regime.

The scaling slope is 1.00 on the interacting stratum, and the regime-dial census resolves the thin-wedge sliver structure (net measure jump 8.9 vs. nominal ± 1884.6).

4.7 The value function as canonical carrier

The tie-break-freeness of the value layer (Proposition 2.6) is directly testable: solve the same parametric problem under two different tie-break conventions and compare layers. $\Phi = c_{\text{bio}}^\top v^*$ is single-valued with no tie-breaking; it is piecewise affine at 4.2×10^{-13} on the flux-event partition, carries one real atom $\Delta\Phi' = -0.006439$ at $t = 0.0358286$, and decouples from the flux-jump hierarchy (value/flux mass ratio 1.7×10^{-6}). Finite-difference shadow prices match Danskin envelope marginals to 6–7 digits. A controlled follower-variable LP reproduces the same structure: the value function stays affine and identical under two lexicographic tie-breaks while the follower’s second-difference mass flips $0.0100 \leftrightarrow 10^{-13}$ — the value layer is canonical (Proposition 2.6), the flux-strain layer is not (Corollary 3.9). The coupling identity of Theorem 2.11 is machine-verified on this same cut: $\Delta\Phi' = c^\top \Delta v'$ with error 0.0 at all 12 events and all 4 clusters, $\Phi = c^\top v^*$ to 10^{-9} , and the 11 invisible events carry flux ℓ_1 slope jumps up to 1.6×10^4 against $|c^\top \Delta v'| \leq 1.51 \times 10^{-7}$ — the decoupling is the objective contraction, not a loss of network information.

5 Empirical association and protein-layer buffering

This section takes the metric to data. Which genes does the network itself flag as rerouting-sensitive, and do those genes respond when the environment actually tightens? The answer comes in layers: first a robust transcriptional association, then an explicit negative result at the protein layer, and finally robustness checks across trajectories, tie-break conventions, and panels.

5.1 The gene panel and GPR classes

Reaction-based sampling on the reference physiology (glucose $5.0 \rightarrow 1.0$, oxygen $22 \rightarrow 5$; iJO1366) leaves 438 of 2,583 reactions active (17.0%) and 435 genes with at least one active reaction. The GPR complexity classes over the active reactions — single-gene 254, isozyme-OR 96, complex-AND 36, mixed 18, no-GPR 34 — are exactly the classes that drive the concavity analysis of Proposition 2.12. Four verification checks were applied to the panel construction. (i) *GPR mapping*: taken from cobra’s authoritative `reaction.genes`, spot-checked against the raw rule strings on four reactions spanning all complexity classes, with set-equality of gene tokens and a bidirectional gene–reaction symmetry check on 50 sampled genes (0 failures over 120 gene–reaction pairs). (ii) *Definition consistency*: the fresh re-derivation reproduces the deposited values exactly (per-gene maxima, the biomass-deficit maximum, and the unmasked/masked panel correlations to four decimals). (iii) *Distinctness*: the panel shares exactly one b-number with the κ_V panel (b2097). (iv) *Non-zero variation*: every panel gene has κ_V variation > 0 over T1–T8 (435/435). For completeness, the geometric metric κ_V is weak at the panel level (r from -0.063 unmasked to $+0.084$ masked); the measure-theoretic κ^μ (§5.3) is the recalibrated predictor on the same panel.

5.2 Multi-condition support

Nine FBA conditions were probed (baseline carbon decline; galactose, trehalose, glycerol-3-P, and proline-N carbon sources; anaerobic-nitrate, oxidative, and osmotic designs): active-reaction support is 438–454 per condition, with union 537 (20.8%), and genes with nonzero κ_V expand $435 \rightarrow 525$ across conditions. The endogeneity check is clean (feed caps non-binding except the designed glycerol-3-P limitation; imposed-burden magnitudes flagged as not interpretable). The rerouting metric is thus well posed across carbon sources and stress designs: its support *expands* with condition diversity rather than migrating — the measure sees more of the network as the environment demands more of it.

5.3 Transcriptional response under carbon depletion

To test whether active-set curvature predicts real biological adaptation, we modeled *E. coli* adapting to carbon starvation on the deterministic lexicographic trajectory (the reference physiology, iJO1366), comparing κ^μ (Definition 2.3) against log-fold carbon-depletion response (M3D (Faith et al., 2008), $n = 424$ genes; Fig 3). The active-set sensitivity metric κ^μ showed a strong, highly significant positive correlation with transcriptional induction: Pearson $r = +0.395$ ($p = 2.6 \times 10^{-17}$), Spearman $\rho_S = +0.414$ (full panel, zeros included). One confound deserves a direct look: because highly expressed metabolic enzymes often show larger absolute dynamic ranges, the raw association could partly reflect baseline expression rather than rerouting; we therefore evaluated a partial correlation controlling for baseline log-phase expression. The association remained highly significant: residualizing the M3D reference level leaves the association intact (partial $r = +0.269$, $p = 1.8 \times 10^{-8}$), and genes in the extreme deciles separate cleanly (top versus bottom response 1.92 vs 0.89 — more than double; MWU 1.3×10^{-7}). The baseline reproduces to the digit ($r = +0.3739$, $n = 433$), and metric invariance holds, $\rho(\kappa^\mu, \kappa_V^{\text{lex}}) = 0.99998$: both metrics sample the same flux-event structure along the same lexicographic trajectory, so the association is an event-structure property of the flux layer — not a metric artifact and not a value-function shadow. A cross-platform replication with the declared metric completes the picture. The carbon-switch arms replicate on the same platform (M3D microarray: glycerol +0.195, acetate +0.166, proline +0.175, switch-MAX +0.206; $n = 424$ genes, all $p < 6 \times 10^{-4}$) but dissociate across platforms — the PRECISE RNA-seq carbon-switch statistic (ten WT non-glucose carbon conditions) gives $r = -0.044$ (partial -0.088 , NS), with individual conditions weakly positive (glycerol +0.126, acetate +0.130, fructose +0.099) except galactose (-0.088). The association is thus specific to the carbon-exhaustion class and the platform-matched design; the cross-platform dissociation is reported as a negative result.

5.4 Anatomy of one switch

The association of §5.3 is statistical; this subsection opens it up and names its parts. The switch is carbon depletion itself — the progressive glucose decline of the reference trajectory — and its anatomy has three layers, each computable from the deposited artifacts.

Which chemistry reroutes. The curvature mass of the sweeps concentrates on the forks of central carbon (§4.2), and the reactions carrying the largest mass are the enzymes of those forks: glucose-6-phosphate isomerase, phosphofructokinase, fructose-bisphosphate aldolase, and the pentose-phosphate entry block. The hexose-phosphate pool, the triose-phosphate pool, the PEP/pyruvate fork, and the anaplerotic TCA forks are where the rerouting of carbon depletion physically happens.

Which regulons the genes belong to. Partitioning the panel by κ^μ quartile and testing the top quartile (110 of the 424 nonzero- κ^μ genes) for enrichment against the PRECISE transcriptional-regulatory annotations (Sastry et al., 2019), with the panel as background, gives a precise answer (Table 2): the global carbon and energy regulons. CRP targets are enriched 2.4-fold, with 56 of the 89 CRP panel targets in the top quartile (Benjamini–Hochberg $q = 8.6 \times 10^{-16}$); Cra, Fur, Fnr, RpoS, Fis, ArcA, NarL, IHF, RpoD, and NagC follow; eleven regulons survive control at $q < 0.10$. The enrichment is induction-specific. CRP’s activation targets carry high curvature (median κ^μ 0.64 against 0.02 for the rest of the panel; Mann–Whitney $p = 0.0016$), while its repression targets do not ($p = 0.43$): induction prices the option set; repression does not track the geometry.

At operon granularity the pattern is stronger. Whole operons sit entirely in the top curvature quartile: the NADH dehydrogenase operon *nuoABCEFGHIJKLMN* (13 of its 13 panel genes; $q = 4.3 \times 10^{-7}$), the ATP synthase operon *atpIBEFHAGDC* (9 of 9), the succinate dehydrogenase operon *sdhCDAB-sucABCD* (8 of 8), cytochrome *bo* oxidase *cyoABCDE* (4 of 4), and *manXYZ*, *ptsHI-crr*, and *pdhR-aceEF-lpd* (3 of 3 each). Within-operon curvature is near-uniform: the mean within-operon $|\Delta\kappa^\mu|$ is 0.044, against 1.045 across operons (381 pairs; permutation $p <$

Table 2. Regulon enrichment of the top- κ^μ quartile (110 of the 424 nonzero- κ^μ panel genes) against PRECISE regulatory annotations, panel background, Fisher exact tests with Benjamini–Hochberg correction across the 31 testable regulons; eleven survive $q < 0.10$ (all shown). The enriched sets are the global carbon, energy, and stress regulons.

regulon	top / panel	fold	p	q (BH)
crp	56 / 89	2.43	2.8×10^{-17}	8.6×10^{-16}
cra	39 / 52	2.89	1.8×10^{-15}	2.8×10^{-14}
fur	24 / 27	3.43	1.6×10^{-12}	1.5×10^{-11}
fnr	46 / 77	2.30	2.0×10^{-12}	1.5×10^{-11}
rpoS	26 / 33	3.04	4.4×10^{-11}	2.7×10^{-10}
fis	22 / 31	2.74	5.6×10^{-8}	2.9×10^{-7}
arcA	41 / 89	1.78	2.6×10^{-6}	1.1×10^{-5}
narL	14 / 20	2.70	2.9×10^{-5}	1.1×10^{-4}
ihf	21 / 38	2.13	5.1×10^{-5}	1.8×10^{-4}
rpoD	86 / 274	1.21	3.1×10^{-4}	9.7×10^{-4}
nagC	6 / 8	2.89	4.8×10^{-3}	1.3×10^{-2}

2×10^{-4}). The control that makes this non-trivial is the GPR structure itself. Restricted to intra-operon pairs catalyzing *disjoint* reactions — genuinely different enzymes, such as the *sdh* gene products against the *suc* gene products — the gap remains 0.082 against 1.045 (205 pairs, permutation $p < 2 \times 10^{-4}$), while pairs sharing a reaction (enzyme-complex subunits; 176 pairs) are trivially identical. Co-transcribed units carry near-uniform curvature across their different enzymes: the option set is priced in pre-packaged units. The co-expression modules that light up in the iModulon annotations are the carbon and energy ones (DhaR/Mlc, Crp-2, ArcA-1, ArcA-2) — the same sector again.

What growth sees of it. Nothing. On all three trajectories of §5.6, the growth-attributed curvature is identically zero — 0 of the 433 panel genes has nonzero κ_c on any path — and the value layer carries between 0.024% and 0.145% of the flux layer’s curvature mass. The association of §5.3 is therefore carried entirely by growth-invisible wall mass. Transcriptional induction during carbon depletion concentrates on the sector where selection on growth is silent: the zero-cost pathway substitutions of Theorem 2.11(i), which is precisely the degeneracy’s design space.

Put together, the anatomy closes a loop that Kochanowski et al. (2013) opened from the other side. They showed that the response to carbon limitation is coordinated largely at the metabolite level, through Crp and local metabolite concentrations; here the regulon that governs the response is the regulon whose operons carry the curvature, and its enzymes sit on the fork metabolites that carry the walls. One chemical corollary follows, as a prediction: of the 237 regulators in the transcriptional network, the sensing molecules of five are themselves metabolites of the reconstruction (FMN, L-tryptophan, adenosylcobalamin, molybdopterin, spermidine); the metabolite-effectors of the enriched regulons should prove to be wall-adjacent pools from Table 1.

Provenance and honesty. The enrichment shares the correlational design and the expression-level confound structure of the parent association (§5.3); the activation/repression split partially controls for direction of effect, but expression-matched per-gene controls are future work, and the PRECISE RNA-seq arms replicate the association only weakly (§5.3), so the magnitude of the enrichment is platform-dependent even where its sign is consistent.

5.5 The layer decision

Which layer carries the association? We answer this by running every value-layer attribution arm under the declared protocol of §5.3 (iJO1366, the reference physiology, $n = 424$ genes). The flux layer κ^μ gives $r = +0.395$, partial $r = +0.269$ ($p = 1.8 \times 10^{-8}$); the c -attribution

arm is supported on the biomass pseudo-reaction alone (0 of 433 genes; Theorem 2.11(ii)); the shadow-price-jump arm gives 51 genes at $r = +0.032$ (partial -0.013 , $p = 0.93$). The value trajectory has *zero* chamber crossings: the path stays in one chamber of Φ throughout — Φ is affine along it, with $d\mu/dq_{\text{glc}} = Y = 0.099544$ ($= |y_{\text{glc}}|$, the constant glucose uptake shadow price; affine intercept -0.0124) and uptake shadow prices jumping by exactly zero — all four value kinks are anchor corners of the trajectory design, of magnitude $Y \Delta(dq_{\text{glc}}/dt)$, carrying no network structure. Value/flux strain mass ratio 1.45×10^{-3} ; the coupling identity holds at machine precision (error 0.0 at every kink). The value-gated flux strain coincides with κ^μ exactly (433/433 genes attain their maximum at a value-kink time), a property of the trajectory design (anchor rerouting dominates per-gene maxima), not evidence of value-layer content. The layer decision: the association lives on the flux layer, with Theorem 2.11 as the exact bridge to the canonical object.

5.6 Path robustness: what survives a trajectory change

To establish that κ^μ captures general network vulnerabilities rather than a quirk of glucose limitation, we tested two alternative environmental stress paths through the same LP (same engine, seed, panel, statistics; only the path and its matched response vary; Fig 4): P0 glucose decline (reference-physiology anchors; the layer-decision control of §5.5), P1 oxygen limitation — progressive transition from aerobic to anaerobic growth ($q_{\text{glc}} = 5$ fixed, q_{O_2} linear from 22 to 1) — and P2 acetate diauxic switch — progressive transition from glucose to acetate ($q_{O_2} = 22$ fixed; glucose $5 \rightarrow 0$ while acetate uptake ramps $0 \rightarrow 10$). Kinks are classified by the chamber test — the parameters are the uptake bounds, so $\nabla\Phi = (y_{\text{glc}}, y_{O_2}, y_{\text{ac}})$: a kink is a *design corner* iff it sits at a trajectory anchor and the uptake shadow prices are continuous across it (jump $\leq 10^{-9}$), and a *chamber crossing* iff they jump.

- (i) The one-chamber property is *path-dependent*: P0 has zero chamber crossings (the §5.5 control, reproduced digit-exactly), while P1 has three genuine crossings ($q_{O_2} \approx 9.3, 6.3, 2.9$; $|\Delta\Phi'|$ up to 0.68; y_{O_2} jumps $+0.032$ at the respiratory boundary) and P2 has four (two large, near the glucose–acetate handoff; two micro-slivers with dual jumps $\sim 10^{-4}$). The single-chamber structure of the reference trajectory is a property of the glucose-decline design, not a law.
- (ii) The layer decision survives the change: on every path the value-layer arms contribute nothing. The c -attribution arm is degenerate (0 of 433 genes — the sparse-objective corollary of Theorem 2.11 is path-free); the chamber-crossing shadow-price arm is null ($r = -0.17$, $n = 68$ on P1; $r = +0.032$, $n = 71$ on P2); value-gated flux strain is a subset of κ^μ (equal or weaker on every path, never stronger).
- (iii) The flux-layer association generalizes: against the matched oxygen response (M3D WT anaerobic vs aerobic contrast) P1 gives $r = +0.318$ (partial $+0.258$, $p = 6 \times 10^{-8}$, $n = 426$); against the matched acetate-switch response P2 gives $r = +0.223$ (partial $+0.160$, $p = 9 \times 10^{-4}$); and the P1/P2 predictors correlate with the primary-panel carbon response at $r = +0.378 / +0.391$ — κ^μ identifies core architectural bottlenecks that dictate transcriptional regulation across multiple environments, and the per-gene ranking is largely trajectory-independent.
- (iv) Theorem 2.11 holds at every kink on all three paths (identity error 0; worst case 1.6×10^{-15}). Value/flux strain mass ratios: 1.5×10^{-3} (P0), 2.4×10^{-4} (P1), 5.5×10^{-4} (P2).

5.7 Tie-break robustness: the association as a property of the protocol class

Because linear programs can possess multiple alternative optimal flux distributions, the flux layer requires a tie-breaking rule — it is selection-dependent by construction (Remark 2.5), and the declared lexicographic tie-break is one member of a family. To confirm that the biological findings do not depend on a specific computational convention, the experiment fixes the entire protocol

Table 3. **Evaluation across alternative tie-breaking protocols** (primary protocol fixed; only the stage-3 selection rule varies). The first-stage growth rate μ and the second-stage parsimonious sum ℓ_1 agree to 0.0 across all variants, confirming the mathematical invariance of the value layer — the measured form of the tie-break-freeness of Proposition 2.6.

Tie-breaking rule	Active genes (n)	Pearson r	Partial r	$\rho_S(\kappa^\mu, \kappa_{\text{TB0}}^\mu)$	Reassigned genes
TB0 (declared)	424	+0.3954	+0.269	—	—
TB1 (fresh seed)	426	+0.3861	+0.270	0.99998	91
TB2 (log-normal)	424	+0.3954	+0.269	0.99999	90
TB3 ($ v $ rule)	425	+0.3959	+0.269	0.99897	109
TB4 (adversarial)	424	+0.3954	+0.269	0.99998	117

of the primary association (iJO1366, reference-physiology anchors, $8\times$ refinement, the 433-gene panel, the M3D carbon-depletion response, the reference-level confound control) and varies *only* the stage-3 tie-break of the engine:

- **TB0 (declared baseline):** $w \sim U(0.5, 1.5)$, seed 20240901, $\min w^\top v$ — the declared metric (control: reproduces the primary association, $r = +0.3954$, digit-exactly);
- **TB1 (seed replication):** $w \sim U(0.5, 1.5)$, seed 20240902 — same family, independent draw;
- **TB2 (dynamic-range expansion):** $w \sim \text{LogN}(0, 1)$ clipped to $[0.05, 20]$, seed 20240903 — a different distribution family with a roughly 130-fold wider dynamic range (support ratio 400:1 versus 3:1);
- **TB3 (absolute-flux rule):** stage-3 objective on the split variables, $\min \sum_r w_r(f_r + r_r)$ — weighted *absolute*-flux selection with TB0’s weights;
- **TB4 (adversarial selection):** $\max w^\top v$ — the far end of the stage-2-pinned optimal face, with TB0’s weights.

Three findings (Table 3, Fig 5). (i) *The association is stable:* across all five rules $r \in [+0.386, +0.396]$ (maximum deviation 0.009; all $p \leq 1.4 \times 10^{-16}$) and the partial correlation is constant to three decimals (+0.269–+0.270). (ii) *The tie-break genuinely binds:* the stage-2-pinned optimal face is multi-vertex, and the alternative rules select different flux vectors at all 57 grid points ($\max \|v_k - v_0\|_\infty$ up to 5.0), changing per-gene κ^μ for 90–117 of 433 genes — yet the per-gene ranking is essentially unchanged ($\rho_S \geq 0.99897$), and the changed values are small on the κ^μ scale (max absolute change 9.3×10^{-5}); the TB1 deviation of 0.009 traces to two boundary genes entering the nonzero support ($n : 424 \rightarrow 426$), not to a reshaping of the metric. (iii) *The value layer is exactly invariant:* stage-1 biomass μ and stage-2 ℓ_1 mass agree across variants to 0.0 — Proposition 2.6 (tie-break-freeness of the value layer) measured on the full trajectory. Together with Corollary 3.9 this completes the robustness picture: the declared selection is a convention, the value layer does not see it, and the flux-layer association is a property of the lexicographic protocol class, not of one rule — a feature of the network’s active-set architecture, not an artifact of solver mechanics.

5.8 Event-measure stabilization (Glivenko–Cantelli type)

The refinement-stabilization question (*do the empirical event point measures of the lex-pFBA map converge as the sampled family grows?*) has a near-term statistical form, executed here on three data sources: random straight cuts through the $(q_{\text{glc}}, q_{O_2})$ signature plane (iML1515, 34×34 census, 350 boundary edges; a pool of 4,000 random cuts yielding 25,107 cut events), the thirteen

parameter sweeps of §4.1 (2 affine, 11 event-bearing), and the panel protocol of this paper (gene panel and trajectory). The metric is the bounded-Lipschitz distance (domains normalized to unit diameter, where it equals W_1 ; exact in one dimension, transport LP in two); each arm carries a permutation null (event locations replaced by uniforms, panel structure kept) and the one-dimensional arms carry the classical Glivenko–Cantelli asymptotic constant $C = \int \sqrt{F(1-F)} dx$ of del Barrio–Giné–Uzet computed from the population (Fig 6).

(i) *Random panels stabilize at the GC rate.* Pooled cut-event locations on the cut coordinate ($d=1$): measured tail slope -0.492 versus null -0.506 , measured/null ratio 1.24 at the largest panel, and the iid prediction $C/\sqrt{n}$ matches the measured curve to 2.5% at the largest panel (mid-range deviations as large as 8.2%); the two-dimensional pooled locations (transport LP) decay consistently with the $d = 2$ rate up to the reference-noise floor. Sweep panels (the thirteen sweep families) stabilize with a heterogeneity penalty that falls from $\approx 1.8\times$ the uniform null at small panels through $\approx 1.5\times$ at $k = 6$ to $\approx 0.5\times$ at $k = 12$, and a tail faster than iid-GC — consistent with the finite-population correction $\sqrt{(13-k)/12}$ of sampling without replacement from thirteen heterogeneous sweeps (measured null-relative ratio 0.49 at $k = 12$, against the pure FPC value 0.29; the residual is the sweep heterogeneity). The gene panel: the κ distribution stabilizes at the GC rate (measured effective constant ≈ 1.6 –2.1 across panel sizes (dipping at $m = 128$), against the population Glivenko–Cantelli constant $C = 2.49$ on the $\log_{10} \kappa^\mu$ axis), and the association itself stabilizes at $r \approx +0.39$ with the Fisher standard error $1/\sqrt{m-3}$.

(ii) *Designed panels stabilize exactly, not statistically.* The trajectory event point measure is a *four-atom* object: the four design corners of the physiology anchors carry the entire flux-layer mass (288.77, the primary association’s flux-layer mass to the digit) — the one-chamber structure of §5.5 in its event form. Anchor-preserving thinning reproduces it *exactly* at every panel size $m \geq 8$ (shape distance 0 to machine precision, $\leq 7 \times 10^{-13}$; total-mass error $\leq 2 \times 10^{-12}$; value-kink census $\equiv 4$); uniform random thinning decays by corner censoring (shape distance $0.12 \rightarrow 0.005$ by $m = 43$, mass error $26\% \rightarrow 0.13\%$ by $m = 29$, machine-zero $(\leq 3 \times 10^{-12})$ from $m = 43$, and exactly 0 at the full panel $m = 57$) — a resolution effect in the sense of Theorem 3.1(iv), not statistical averaging.

Interpretation: the mean-field intuition is confirmed in its statistical form — the empirical event measures over random panels (cuts, sweeps, genes) are Glivenko–Cantelli objects — and simultaneously sharpened: stabilization has two regimes, statistical for random panels and structural (exact, design-immediate) for designed panels. The deterministic refinement-sequence form of the question (refinement along explicit model-family sequences) remains open.

5.9 Platform-by-class disambiguation (2×2)

Is the association a carbon-depletion artifact, or a platform artifact? The question has a clean 2×2 answer: two platforms crossed with two perturbation classes, executed with the geometric κ_V metric (rank-equivalent to κ^μ at $\rho = 0.99998$). Carbon switch on microarray (M3D glycerol/acetate/proline vs glucose) switch-MAX $r = +0.196$; stress on microarray (heat shock, acid shock, ciprofloxacin) MAX $r = +0.298$; carbohydrate-responsive transcript levels: carbon starvation on RNA-seq (GSE64021 (Barrett et al., 2013) time course) MAX $r = +0.191$; heat shock on RNA-seq MAX $r = +0.345$ ($n = 241$ –433; permutation $p \leq 0.003$). Signed responses flip sign (stress arms -0.22 to -0.32): the metric tracks *rerouting magnitude*, not direction, on both platforms and both classes. The association is thus neither platform-specific nor carbon-class-specific.

5.10 Protein abundance versus protein change

Does the transcriptional response propagate downstream to change protein levels? We evaluated two quantitative proteomic datasets for *E. coli* — baseline protein abundance averages from the

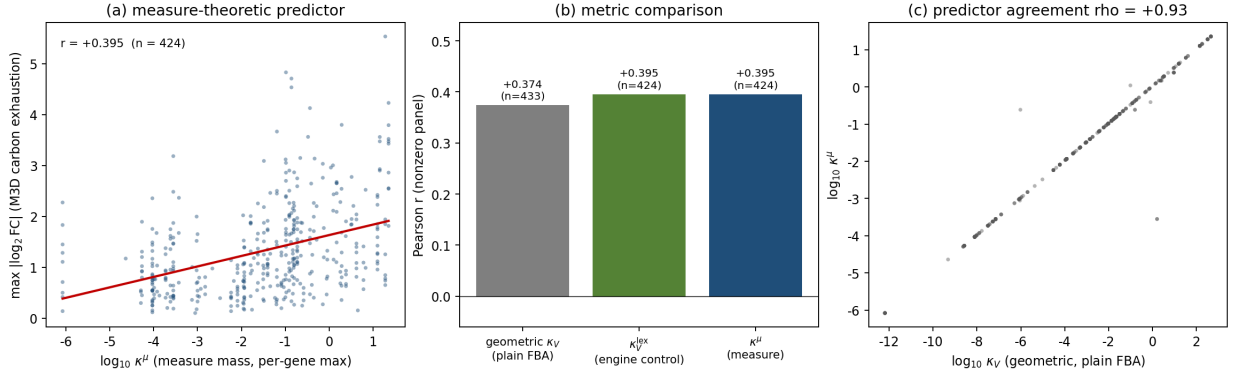


Fig 3. The primary association. (a) Per-gene $\log_{10} \kappa^\mu$ (flux-layer measure mass, per-gene max over the gene’s reactions) versus M3D carbon-depletion response ($\max |\log_2 \text{FC}|$, four stationary contrasts, $n = 424$ nonzero genes); (b) metric comparison: the geometric time-course κ_V (reference physiology, plain-FBA), the engine control κ_V^{lex} , and the measure-theoretic κ^μ on the same trajectory; (c) predictor agreement between κ^μ and the plain-FBA κ_V .

PaxDb integrated database (Wang et al., 2015), and condition-dependent quantitative proteomics across the GSE64021 (Barrett et al., 2013) starvation time courses ($n = 169\text{--}429$) — and our analysis revealed a striking biological dissociation. First, abundance *level* associates (PaxDb baseline $r = +0.334$): high-rerouting genes are abundant proteins. Second, abundance *change* does not: in sharp contrast to mRNA, dynamic protein log-fold change showed no correlation with κ^μ ($r = +0.008$ to $+0.032$), and the transcript–protein fold-change coupling across matched conditions was essentially zero ($r = +0.020$, $n = 799$). The transcriptional association thus stops at translation, consistent with buffering: rerouting capacity is carried at the transcript level, and the translational cost is deferred until a rerouting bottleneck actually demands it (§5.11).

5.11 Protein-layer null

The decisive negative result comes from condition-dependent quantitative proteomics (Schmidt et al. (2016); 22 conditions, triplicates, $n = 366$): in this multi-condition dataset, the correlation between κ^μ and protein fold change was weakly negative ($r = -0.083$), directly opposing the strong transcriptional correlation ($r = +0.419$) observed for the same genes — the association that is robust at the transcript layer vanishes entirely at the protein layer. Transcription does not propagate through translation; buffering at translation is the consistent model, positioning Kochanowski et al. (2013) as prior art for the protein-layer null.

6 Discussion

6.1 Unifying discrete switches and continuous models

For decades, mathematical modeling in systems biology has been divided between two distinct approaches: continuous, smooth ordinary differential equation (ODE) models of enzyme kinetics, and discrete, piecewise-linear optimization models such as FBA. Our work bridges this divide by providing an explicit geometric framework for the second derivatives of constraint-based models: the sensitivity of a metabolic network is captured by an atomic Radon measure concentrated on the active-set boundary complex. The sensitivity objects of the framework are one measure at different resolutions — the atomic measure of Definition 2.1, its coarse-grainings $\mu * \phi_\sigma$, and its integrals along parameter trajectories (κ^μ). Through the refinement–resolution bridge (Theorem 3.1), discrete active-set transitions converge weakly to continuous curvature densities under spatial averaging, but total-variation measures do not converge due to triangulation anisotropy. A smooth model of metabolic sensitivity is valid only when observed through an intermediate

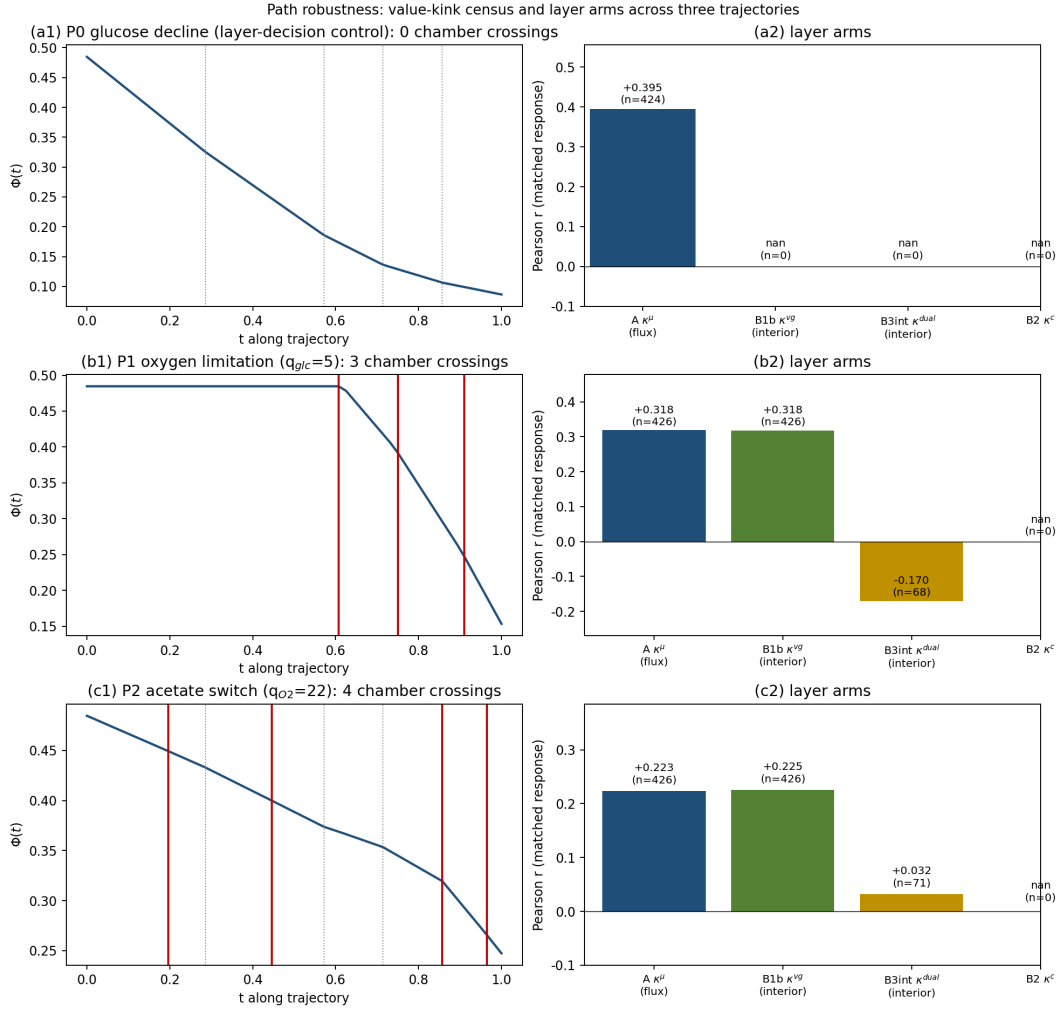


Fig 4. Path robustness. Three trajectories through the same LP (P0 glucose decline, P1 oxygen limitation, P2 acetate switch) with matched M3D responses: the one-chamber property is path-dependent (P1: 3 chamber crossings; P2: 4), while the layer decision and the flux-layer association survive on every path; the P1/P2 predictors correlate with the primary-panel carbon response at $r = +0.378/+0.391$.

909 resolution window ($h \ll \sigma \ll L_{\text{var}}$); the $\sigma \rightarrow 0$ limit is the atomic object, and outside the window
 910 — σ below the mesh scale or above the variation scale — the smooth representation breaks down,
 911 the true physical behavior of the network being inherently discrete and linear, characterized by
 912 $O(\varepsilon)$ holonomy scaling. Two items remain open: Conjecture 3.6 (general complex stability) and
 913 second differences on measured time courses (under the (ε, σ) design law). The statistical form
 914 of the stabilization question is settled here (§5.8): random panels are Glivenko–Cantelli objects
 915 and designed panels reproduce the event measure exactly; the deterministic refinement-sequence
 916 form (refinement along explicit model-family sequences) remains open.

917 6.2 Relation to metabolic control analysis

918 Metabolic control analysis (Kacser and Burns, 1973; Heinrich and Rapoport, 1974) asks a sister
 919 question with different machinery: how much does a steady state move when one enzyme’s kinetics
 920 change? Its answers — flux and concentration control coefficients, elasticity coefficients — are
 921 *local smooth sensitivities*: first derivatives at a single stable steady state of a kinetic model, defined
 922 exactly where the response is differentiable. The objects of this paper are their complements. κ^{μ}
 923 is a global, path-integrated, second-order measure concentrated precisely where differentiability

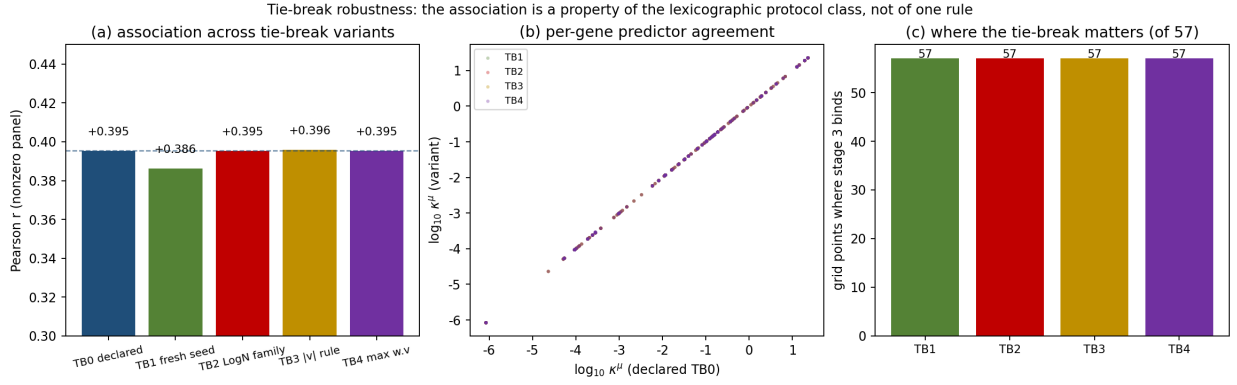


Fig 5. Tie-break robustness. (a) The association across the five stage-3 selection rules (declared plus four variants); (b) per-gene predictor agreement between the declared rule and each variant; (c) the number of grid points at which the tie-break binds (of 57). The value layer is invariant to 0.0 across all variants.

fails — on active-set boundaries — and it needs only stoichiometry and bounds, not kinetic parameters. The two frameworks meet in the middle of the resolution window: the σ -smoothed curvature densities of Theorem 3.1 are the regime in which smooth sensitivity language is valid, and there the averaged discrete measure plays the role that control analysis assigns to smooth response coefficients. Outside the window — at a wall — the steady state is not unique or the response is kinked, control coefficients are undefined, and the discrete measure is exactly at home.

The structural analogies are exact enough to be worth naming. The flux-control summation theorem ($\sum_i C_i = 1$) says that the system-level view sees a fixed contraction of the enzyme layer; the coupling identity $D^2\Phi = \sum_r c_r D^2v_r^*$ (Theorem 2.11) is the parametric-LP form of the same statement, and the invisible events of Theorem 2.11(i) are the degenerate faces where a control coefficient would have to divide its response among infinitely many co-optimal states. The empirical dissociation adds a sharper point: control analysis lives at the enzyme level — kinetic parameters set the response — whereas the transcriptional association here lives on rerouting geometry that the protein layer does not carry (§5.11). Regulation tracks the geometry of the option set, not the kinetic magnitudes of the enzymes that implement it.

6.3 Why growth rate fails to predict gene expression

A key conceptual takeaway of our work is the value–flux coupling theorem (Theorem 2.11). In systems biology, metabolic stress is frequently evaluated through growth rate reductions or dual shadow prices. However, the coupling identity proves that the objective value function acts as a mathematical contraction: it sums internal reaction curvatures weighted by objective coefficients ($D^2\Phi = \sum_r c_r D^2v_r^*$), grounding the association’s flux-layer footing and complementing the measured metric invariance of §5.3. Because cells routinely exploit alternative pathways of equal growth during nutrient transitions, most metabolic rerouting is objective-invisible (the zero-cost substitutions of Theorem 2.11(i)), and when the objective is biomass production the contraction zeroes out all internal pathway adjustments that do not reduce immediate growth: in our *E. coli* simulations, 11 out of 12 internal rerouting events were entirely silent in the growth rate. Consequently, transcriptional regulation correlates strongly with the internal flux metric κ^μ ($r = +0.395$) but has no correlation with value-layer derivatives (shadow-price arm $r = +0.032$, $p = 0.93$, $n = 51$). Regulatory networks are organized to manage internal pathway capacity and prevent local traffic jams, not simply to track external growth yield: transcriptional response tracks rerouting, and rerouting is a flux-layer quantity.

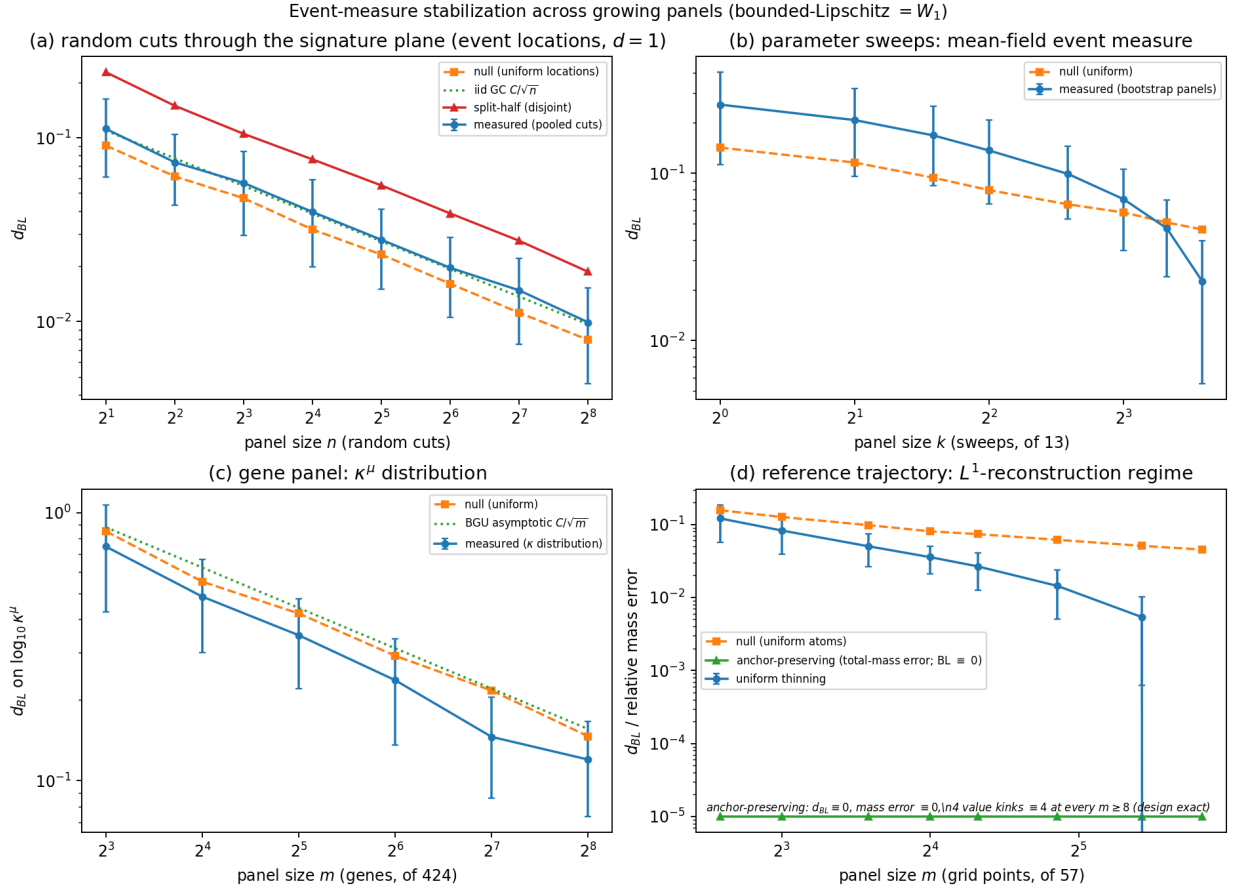


Fig 6. Event-measure stabilization. (a) Random-cut panels: pooled event-location distance decays at the Glivenko–Cantelli rate (measured tail slope -0.492 ; null -0.506 ; iid prediction $C/\sqrt{n}$ to 2.5% at the largest panel, mid-range deviations as large as 8.2%); (b) Parameter-sweep panels: stabilization with a finite-population-corrected, heterogeneity-penalized tail (null-relative ratio 0.49 at $k = 12$ against the pure FPC 0.29); (c) Gene panel: the κ distribution at the GC rate (effective constant ≈ 1.6 – 2.1 , dipping at $m = 128$, vs. the population constant $C = 2.49$); (d) Reference trajectory: the L^1 -reconstruction regime — random thinning decays by corner censoring, anchor-preserving thinning is exact ($d_{BL} \equiv 0$) from $m \geq 8$.

6.4 Biological implications of translational buffering

Why transcribe a rerouting program without translating it? The decoupling between transcriptional induction and protein synthesis observed in our study highlights an elegant cellular survival strategy. Synthesizing large, multi-subunit metabolic enzymes is energetically expensive, and when nutrients become scarce, a cell under catabolic limitation cannot afford to blindly translate every enzyme it might need. By upregulating mRNA transcripts for high- κ^μ bottleneck pathways, the cell pre-positions the necessary genetic templates: if an active-set boundary is crossed and a metabolic pathway becomes completely blocked, local translation can be released immediately; if the starvation condition resolves, the untranslated mRNA can be recycled at minimal metabolic cost. The measured dissociation — transcript $+0.419$ against protein $r = -0.083$ on the same 366 genes (§5.11) — is the quantitative signature of this standby model, for which Kochanowski et al. (2013) is the prior art at the metabolite level.

6.5 Where metabolic memory lives

Two established results, put side by side, force a structural conclusion. First, 66% of closed genotype cycles retain flux-state drift after full restoration (§4): the flux state after an $A \rightarrow$

$AB \rightarrow B \rightarrow A$ cycle is not the initial state, with a median residual drift of $\sim 110 \ell_1$ units. Second, the protein layer does not carry the association: protein fold change explains at most 0.7% of the curvature variance ($r = -0.083$, $R^2 = 0.007$, $n = 366$; §5.11), while transcripts on the same genes carry $r = +0.420$; under the locked κ^μ metric the picture is unchanged (protein $r = -0.098$, $p = 0.06$; transcript $r = +0.339$). A state difference of median size 110 cannot be stored in a layer that moves by less than one percent of the signal: enzyme abundance is quantitatively excluded as the carrier of metabolic hysteresis. Transcripts carry the association but are instructions, not state — the menu, not the meal. The persistent state variable must therefore be fast and post-translational: metabolite pools and their modification states.

This deduction tells the wet-lab investigator where not to look (enzyme titers) and where to look. Kochanowski et al. (2013) document metabolite-level passive coordination on precisely this class of timescales, and the wall coordinates of §4.2 name the candidate pools: the F6P/GAP/PEP forks that carry most of the curvature mass. A testable prediction follows. In public metabolomics time series across nutrient transitions, branch-metabolite pools should show slope kinks at wall crossings while dedicated-metabolite pools respond smoothly; and the same pools, not dedicated ones, should carry the non-reverting offset after a restoration cycle. The branch-metabolite table is thus the shared substrate of two of this paper’s claims — the chemical coordinates of the walls and the carrier of their memory.

Limitations.

- 1. Design and scope.** The study is correlational by design, and its scope is a single organism; whether the protein-layer buffering null extends beyond *E. coli* is open.
- 2. Operational active sets.** The active sets used throughout are operational (obtained from solved LPs) rather than combinatorial.
- 3. Selection dependence.** The lexicographic tie-breaking is an engine convention: the value-function carrier is immune to it, the flux layer declares it (Remark 2.5), and the association is stable across all five selection rules tested (declared plus four variants, §5.7).
- 4. Path dependence.** The one-chamber structure of the reference path is condition-specific (§5.6), while the layer decision itself is path-robust.
- 5. Cross-platform generality.** The cross-platform carbon-switch dissociation (PRECISE arm, §5.3) bounds the generality of the association.
- 6. Census sensitivity.** The active-reaction census of the reference physiology (438/2,583 and the GPR class counts) is plain-FBA and solver-version-sensitive at the ± 1 -reaction level, while the gene-level panel (435) is stable (Appendix A).
- 7. Stabilization limits.** The sweep arm of the stabilization experiment is panel-limited (thirteen sweep families, two of them affine — the finite-population correction is part of the reported result), and the two-dimensional arm is floor-limited at large panels by the transport-LP atom cap.
- 8. Enrichment and prediction provenance.** The regulon and operon enrichment of §5.4 is correlational and shares the parent association’s confound structure; the metabolomics, cross-species, and effector statements of §§4.2–6.5 are predictions, labeled as such and untested here.

The companion theory paper. The categorical and homotopy-theoretic reading behind the active-set curvature interpretation — the 2-category $\mathbf{StCon}(B)$ of stratified connections with its gluing theorem, the Fisher-minimal transport law on constant-active-set strata, the optic-composition semantics with its contraction analysis, the filtered-colimit construction of autocatalytic sets, the viability-kernel reading of its dynamical closure test (a finite-time, feedback-certifying probe of kernel and capture-basin membership, with the Poincaré/averaging analysis

of its pathwise levels), and the homotopy-type-theoretic extension — is developed in full, as a standalone theory paper with its own proofs, machine verifications, and per-result proof-status marking (the homotopy-type-theoretic extension is proof-sketch-marked there; the stratified gluing is descent-by-citation with construction-level verification), in a companion manuscript (Abae, 2026). The division of labor is deliberate: this paper is the application — the measure-theoretic curvature of parametric FBA, the metric κ^μ , and the genome-scale empirical association with its layer decision — and carries only a brief adapted statement of the load-bearing definitions (§2.6); the companion carries the framework. Neither manuscript depends on the other’s claims.

Scope of the contribution. The mathematics is classical: parametric linear programming (chamber complexes, active sets, Danskin duality (Danskin, 1967)), Alexandrov’s theorem (Alexandrov, 1939), and distributional second derivatives of piecewise-affine maps; Theorem 2.11 is an elementary identity. The contribution is the systematic *application* of this machinery to flux-balance models (Varma and Palsson, 1994; Orth et al., 2010; Lewis et al., 2010) — the two-layer measure structure of the value/flux pair, the exact coupling identity, the signed extension to arbitrary GPR structure — together with the empirical association and its layer decision. No new measure-theoretic theorem is claimed; the nearest prior art (phenotype phase planes (Edwards et al., 2001; Ibarra et al., 2002), mpLP value-function theory (Borrelli et al., 2003), discrete Monge–Ampère (Gutiérrez, 2001), Regge-type defect measures (Regge, 1961; Cheeger et al., 1984)) is cited where it applies.

7 Methods

7.1 The lexicographic pFBA engine

To ensure that optimal flux trajectories are unique, deterministic, continuous, and piecewise linear, all parametric linear programs were solved using a purpose-built three-stage lexicographic solver — the engine that makes every preceding experiment reproducible — on the variable split $x = (v, f, r)$ with $v = f - r$, $f, r \geq 0$, stoichiometry $Sv = 0$, and flux bounds $\ell(\theta) \leq v \leq u(\theta)$ entering affinely through the uptake bounds. The stages are strict lexicographic optimizations, each pinned at the previous stage’s optimum:

$$\begin{aligned} (1) \quad & \mu^* = \max c_{\text{bio}}^\top v && \text{(growth)} \\ (2) \quad & s^2 = \min \sum_r (f_r + r_r) \quad \text{s.t. } v_{\text{bio}} \geq \mu^*(1 - \varepsilon_\mu) && \text{(parsimony)} \\ (3) \quad & \min w^\top v \quad \text{s.t. } \sum_r (f_r + r_r) \leq s^2(1 + \varepsilon_s) && \text{(tie-break)} \end{aligned}$$

with relaxation tolerances $\varepsilon_\mu = \varepsilon_s = 10^{-9}$; the weight vector $w \in \mathbb{R}^m$ was sampled from $w \sim U(0.5, 1.5)$ using fixed seed 20240901 and held fixed thereafter. Stage 3 exists for a reason: plain pFBA is vertex-degenerate on these models, with repeated warm-started solves of the same LP flipping 0.69 reactions on average between vertices of the optimal face. The LPs are solved with HiGHS (Huangfu and Hall, 2018) via `scipy.optimize.linprog`, cold-started and deterministic; a presolve-disabled retry handles the rare infeasibility mis-report. The lex optimum exists, is unique by construction, and is continuous piecewise affine in θ (Lemma 2.10); the tie-break is part of the metric’s definition (Remark 2.5), and its robustness is measured in §5.7. The active-reaction census of the reference physiology is the one plain-FBA computation retained (kept for comparability with the geometric metric κ_V); it is solver-version-sensitive at the ± 1 -reaction level, while the gene-level panel is stable.

7.2 Models and data provenance

Models. The genome-scale models *E. coli* iJO1366 (2,583 reactions) (Orth et al., 2011) and iML1515 (Monk et al., 2017) were retrieved from the BiGG database — iJO1366 for the associa-

tion experiments of §5, iML1515 for the computational batteries of §4 and the value–flux coupling verification cut — and are loaded from the local BiGG JSON copies with SHA-256 manifests.

M3D microarray compendium. Transcriptional profiles were obtained from the Many Microbe Microarrays Database (M3D) *E. coli* compendium v4 Build 6 (Faith et al., 2008, 2007), using uniformly normalized arrays: 4,297 gene probes $\times$ 907 arrays, $\log_2$ uniformly normalized. The carbon-depletion response is the maximum $|\log_2 \text{FC}|$ over the four stationary time points (135/330/480/720 min) against the MOPS glucose reference; the oxygen-limitation and acetate-switch responses use the M9 WT anaerobic versus aerobic and MOPS acetate versus glucose contrasts. The reference-level confound control is the mean log-phase WT MOPS glucose expression. The 117 MB compendium tarball is not committed (repository size); it is re-downloadable from the recorded URL with the committed SHA-256 (e73088f7...).

PRECISE. The cross-platform arm uses the ten WT non-glucose carbon conditions of the PRECISE compendium (Sasthy et al., 2019) (RNA-seq, TPM). PRECISE has no carbon-depletion class; the dissociation is reported as a negative result with per-condition values (§5.3).

Proteomics. Protein copy numbers were obtained from the PaxDb integrated database (Wang et al., 2015), and condition-dependent quantitative proteomic measurements across 22 experimental conditions were obtained from Schmidt et al. (2016) (triplicates); the GSE64021 (Barrett et al., 2013) time course supplies the RNA-seq carbon-starvation and the quantitative proteomics arms (§§5.9–5.10). Gene–Protein–Reaction (GPR) associations were parsed using COBRApy, classifying each active reaction into single-gene, isoenzyme OR-branches, multi-protein AND-complexes, or mixed architectures.

7.3 Panel construction and GPR association

The gene panel is built by reaction-based sampling: solve the lexicographic trajectory on the reference physiology (glucose 5.0 $\rightarrow$ 1.0, oxygen 22 $\rightarrow$ 5; eight anchors, 8 $\times$ per-segment trajectory refinement), take the 438 reactions active ($|v| > 10^{-9}$ at any grid point) on the plain-FBA census of the reference physiology, classify each active reaction’s GPR rule (single-gene 254, isozyme-OR 96, complex-AND 36, mixed 18, no-GPR 34), and assign each gene the maximum over its associated reactions. The expression panel consists of the 433 genes of this set carrying M3D response data; the primary correlation uses the 424 genes with nonzero κ^μ . The near-colliding gene/reaction counts (424/433/435/438/440/525/537) are disambiguated in Appendix A.

7.4 Statistical protocols

The predictor variable was evaluated as $\log_{10}$ of the per-gene κ^μ , reflecting the log-linear distribution of the metric across five orders of magnitude; the experimental response variable was the maximum absolute $\log_2$ -fold change, $\max |\log_2 \text{FC}|$ of the matched contrast. Reported statistical metrics include two-sided Pearson correlation coefficients (r) with analytical p -values, Spearman rank coefficients (ρ_S ; raw predictor ranks, full panel and nonzero subset as labeled), partial correlations controlling for baseline log-phase expression (the M3D reference level, by linear residualization of both variables), permutation tests (10^5 random label shuffles), and bootstrap 95% confidence intervals (10^4 resamples); decile differences were tested using one-sided Mann–Whitney U tests. All tests are two-sided except the decile contrast. The PRECISE per-condition arms use the same protocol with TPM log-fold changes. The enrichment statistics of §5.4 use Fisher exact tests of the top- κ^μ quartile against PRECISE regulatory annotations with the panel as background, Benjamini–Hochberg corrected across the testable sets (31 regulons, 31 operons, 18 iModulons); the within-versus-across-operon curvature gap and its disjoint-reaction GPR control use 5,000-fold label permutation (p reported as $< 2 \times 10^{-4}$). The wall-coordinate, class-share,

and order-rule statistics of §§4.2–4.5 aggregate the deposited per-reaction curvature masses of the eleven boundary-crossing M1 sweeps (each normalized by its own total) and the deposited M3 double-mutant panels, with 10^4 -fold class-label permutation for the class shares. The worked example of §2.5 is machine-verified by LP re-solves at wall-straddling parameter values under both tie-break variants. Every manuscript number traces to a deposited artifact file and is re-derived by the automated numeric verification described under Reproducibility (344 checks).

7.5 Reproducibility

All experiments deposit their result files and generating scripts in the public repository cited in the Data Availability statement. Independent-seed reproductions are recorded for the refinement prototype and the stress batteries. The layer-decision, path-robustness, and tie-break-robustness analyses, the cross-platform arm, and the event-measure stabilization re-run in minutes each on a single workstation. An automated suite of 344 numeric checks re-derives every manuscript number from the deposited artifacts and is re-runnable end-to-end.

A Disambiguation of near-colliding counts

Near-colliding counts in the manuscript, each traceable to a single deposited artifact file (Data Availability):

Value	Meaning (source)
2,583	iJO1366 reactions (model file; recomputed)
438	reactions active on the reference physiology, 17.0% (plain-FBA census, ± 1 solver sensitivity)
440	reactions with $D_2 > 10^{-8}$ along the reference trajectory (reaction-level event count, deposited event file)
433	panel genes with M3D response (panel file row count)
424	panel genes with nonzero κ^μ — the n of the primary correlation (deposited metric file)
426	nonzero genes on the P1/P2 paths (deposited path file)
435	genes with ≥ 1 active reaction, reference physiology (census file; recomputed exactly)
1113 454	max active reactions across the nine multi-condition panels (deposited multi-condition results)
537	union of active reactions across conditions, 20.8% (deposited multi-condition results)
525	genes with nonzero κ across conditions (deposited multi-condition results)
4	trajectory atoms of the event measure — the four design corners carrying the full flux-layer mass 288.77; identical to the value-kink positions (deposited event files)
350	boundary edges of the $(q_{\text{glc}}, q_{O_2})$ signature census, full 34×34 grid (deposited census grid)
25,107	cut events in the random-cut pool, 4,000 cuts (stabilization artifact)
1,516	iML1515 genes, single knockouts (deposited knockout file)
2,779	double-knockout pairs, five panels (deposited knockout file)

B Proofs

This appendix proves the results of the manuscript in full. Where an argument would repeat textbook material, the standard source is cited in lieu of reproduction; the two long technical arguments (Lemma 2.10 and the reconstruction part (iv) of Theorem 3.1) are delegated to Appendix C and cited there.

B.1 Proof of Proposition 2.6

Write the primal as $\max\{c^\top v : Sv = 0, \ell(\theta) \leq v \leq u(\theta)\}$ with ℓ, u affine in θ . Its dual is

$$\min \{u(\theta)^\top w^+ - \ell(\theta)^\top w^- : S^\top y + w^+ - w^- = c, w^\pm \geq 0\},$$

with y free. The constraint set $\mathcal{D}$ of the dual does not involve θ , so by weak duality every dual-feasible triple furnishes an affine upper bound on Φ , and by strong duality (feasibility and boundedness are hypotheses) the value equals the dual optimum. The dual objective is linear in (w^+, w^-) for each fixed θ , so the optimum over the polyhedron $\mathcal{D}$ is attained at one of its

finitely many vertices. Hence Φ is the pointwise minimum of finitely many affine functions of θ and is therefore concave and piecewise affine (the polyhedral value-function structure is classical (Rockafellar, 1970; Borrelli et al., 2003)).

Since Φ is concave piecewise affine, $-\Phi$ is convex piecewise affine, and for every continuous piecewise-affine function the distributional second derivative is the finite signed facet measure computed in the proof of Proposition 2.7 below; each facet density of $D^2(-\Phi)$ is positive semidefinite because $-\Phi$ is convex — the jump of the gradient across a facet of a continuous convex piecewise-affine function is purely normal, $[\nabla(-\Phi)]_F = \alpha_F n_F$ with $\alpha_F \geq 0$ by monotonicity of the subdifferential. This is the piecewise-affine instance of the Alexandrov Hessian measure of a convex function (Alexandrov, 1939); in particular $D^2(-\Phi)$ is a finite symmetric positive-semidefinite matrix-valued Radon measure. As a measure it is a function of Φ alone, and Φ is a function of the LP data (c, S, ℓ, u) alone: no optimal-flux selection rule enters either object. Every measurable selection v^* of optima reproduces the same $\Phi = c^\top v^*$, so $D^2(-\Phi)$ is selection-invariant; the selection acts only on the flux layer $D^2 v^*$, whose mass moves by ten orders of magnitude between two lexicographic tie-breaks while the value layer stays exactly invariant (the controlled follower-variable experiment, §4.1).

For the GPR structure, recall that under multiplicative activity scaling the capacity of a GPR-constrained reaction is built from gene activities by min (AND) and max (OR) nodes. *AND-only rules*: the capacity $\min_k a_k(\theta) u_j$ is the conjunction of the affine inequalities $v_j \leq a_k(\theta) u_j$, one per participating gene, so the feasible set remains a polyhedron with θ entering only through affine right-hand sides, and the concavity argument above applies verbatim (AND-only subnetworks retain concavity). *Exchange-bound families*: the bounds are affine in θ by assumption; the same argument applies. *Single-gene axes*: hold all activities at the reference level and vary one gene. AND gates fall under the previous case. For an OR gate with further genes held at reference activity, the branch $\max(a_g(t), 1) u_j$ is dominated by the constant reference branch for $a_g(t) \leq 1$ — the knockdown direction of the empirical trajectories — so the bound is constant; if the varying gene is the gate’s only member, the bound is affine. Either way the axis restriction is a parametric LP with affine or constant bounds, hence concave. *Failure under OR*: an OR gate with two genes, capacity $\max(\theta_1, \theta_2)$, all else unbounded, gives $\Phi(\theta_1, \theta_2) = \max(\theta_1, \theta_2)$, which is convex and concave only if affine: at $\theta_A = (1, 0)$ and $\theta_B = (0, 1)$ the midpoint $(\frac{1}{2}, \frac{1}{2})$ violates concavity by exactly $\frac{1}{2}$ (the canonical measured violation; the random off-diagonal two-gene probe violates by 0.548). *OR-containing rules*: the value function is the maximum of finitely many concave piecewise-affine functions, one per branch selection, hence continuous piecewise affine, and it carries the signed crease measure of Proposition 2.12; only the Monge–Ampère layer (Propositions 2.7 and 2.8) requires the concave class. $\square$

B.2 Proof of Proposition 2.7

(i). Let $\{\tau\}$ be the finite cell decomposition on which f is affine, with cell gradients $g_\tau = \nabla f|_\tau$. For a test function $\varphi \in C_c^\infty(\Omega)$ and indices i, j , the definition of distributional derivatives and a cellwise Green identity give

$$\langle \partial_i \partial_j f, \varphi \rangle = \int_\Omega f \partial_i \partial_j \varphi \, \text{dvol} = - \sum_\tau (g_\tau)_j \int_{\partial\tau} \varphi (\nu_\tau)_i \, \text{d}\mathcal{H}^{p-1},$$

where ν_τ is the outward normal of τ ; the boundary terms on $\partial\Omega$ vanish for φ compactly supported. On an interior facet F shared by cells τ^+, τ^- the two outward normals are opposite, so the facet contributes $-([g]_F)_j (n_F)_i \int_F \varphi \, \text{d}\mathcal{H}^{p-1}$ with $[g]_F = g_{\tau^+} - g_{\tau^-}$ and $n_F = \nu_{\tau^+}$; the jump is constant along F because f is affine on each cell. Since f is continuous, the tangential components of the gradient agree across F (both equal the tangential derivative of the common trace), so $[g]_F$ is purely normal, $[g]_F = [\partial_n f]_F n_F$, and the facet density $(g^+ - g^-) \otimes n_F = [\partial_n f]_F n_F \otimes n_F$ is symmetric and constant on F . Hence the singular part of $D^2 f$ is the finite facet sum $\sum_F [\partial_n f]_F n_F \otimes n_F \mathcal{H}^{p-1}|_F$. There are no atoms: each facet F incident on a codimension-two

vertex v meets $B_\varepsilon(v)$ in $(p-1)$ -measure at most $C\varepsilon^{p-1}$ (a ball of F 's affine hull), and there are finitely many facets, so the total variation mass in $B_\varepsilon(v)$ is $O(\varepsilon^{p-1}) = O(\varepsilon)$ in the plane — the measured log-log slope 1.000.

(ii). For convex f the Monge–Ampère measure is defined by $\nu_f(E) = |\partial f(E)|$, the Lebesgue measure of the image of the subdifferential (Gutiérrez, 2001). At an interior cell point the subdifferential is the single gradient; at a facet point it is the segment $\text{conv}\{g^+, g^-\}$, of area zero; at a codimension-two vertex v the subdifferential is the convex hull of the gradients of the affine pieces active at v (the max rule for subdifferentials, Rockafellar, 1970), which for three or more active pieces in the plane is a two-dimensional polytope — the normal fan of v . Hence $\nu_f(\{v\}) = |\text{conv}\{n_i\}|$, the volume of the normal fan, and these are the only atoms, since facet-level subdifferentials carry zero two-dimensional area. The measured constancy of the atom under ε -shrinking of the vertex neighborhood is the $\varepsilon \rightarrow 0$ limit of this identity (0.5 for the canonical right-triangle fan). $\square$

B.3 Proof of Proposition 2.8

Let $g = -\Phi$, convex piecewise affine (Proposition 2.6). The minimum of a linear objective over the dual polyhedron $\mathcal{D}$ is attained at a vertex, so g is the maximum of the finitely many affine dual objectives $C_i(\theta) = -(u(\theta)^\top w_i^+ - \ell(\theta)^\top w_i^-)$ indexed by the basic dual solutions $(\lambda_i, y_i, w_i^\pm)$; with the parameters carried into the upper bounds as $u(\theta) = u_0 + U\theta$, the gradient of the i -th piece is $\nabla C_i = -U^\top y_i$ (the bound prices y_i are the sensitivity coefficients of the value function — Danskin's theorem (Danskin, 1967)). The pieces active at θ_v are exactly those of the basic dual solutions optimal at θ_v , and the dual optimal face is the face of $\mathcal{D}$ on which the θ_v -objective attains the optimum, whose vertices are precisely those basic optimal solutions (Ziegler, 1995; Borrelli et al., 2003).

By the max rule (Rockafellar, 1970), $\partial g(\theta_v) = \text{conv}\{\nabla C_i : i \text{ active}\}$, the normal fan of g at θ_v ; and by the subdifferential-image formula for the Monge–Ampère measure (proof of Proposition 2.7(ii); Gutiérrez, 2001),

$$\det D^2(-\Phi)(\{\theta_v\}) = \nu_g(\{\theta_v\}) = |\partial g(\theta_v)| = |\text{conv}\{\nabla C_i\}|,$$

the area of the normal fan: the first identity of the display. For the second, the linear map $y \mapsto U^\top y$ sends the dual optimal face to the polytope $\text{conv}\{U^\top y_i\} = -\text{conv}\{\nabla C_i\}$ (linear maps commute with convex hulls), so the two polygons are reflections of each other through the origin and have equal area:

$$|\text{conv}\{\nabla C_i\}| = |\{U^\top y : (\lambda, y) \text{ optimal dual}\}|.$$

The machine verification enumerates the dual optimal face directly: vertex refined to machine precision by the exact triple-tie solve, $\Phi(\theta_v)$ matching the LP value to 10^{-15} , fan area 0.235 against dual-face enumeration area 0.235000068 over 16 direction LPs. $\square$

B.4 Proof of Lemma 2.10

The proof is delegated to Appendix C: well-definedness of the staged selections by compactness and coordinate pinning; polyhedral chamber structure of each stage by multi-parametric LP theory (Borrelli et al., 2003); polyhedrality of the coordinate-fixing step by Fourier–Motzkin elimination (Ziegler, 1995); continuity and piecewise affinity of the inverse of the injective polyhedral projection. $\square$

B.5 Proof of Theorem 2.11

$\Phi = c^\top v^*$ holds pointwise by optimality. Both sides are continuous piecewise affine (Lemma 2.10 for v^* ; LP value function theory for Φ), and differentiation of distributions is linear: applying D^2 to the pointwise identity gives the measure identity $D^2\Phi = \sum_r c_r D^2 v_r^*$. Restricting to a crease

1215 facet F gives the per-crease form $\Delta\Phi'(t_k) = c^\top \Delta v'(t_k)$ at every crease time t_k of any piecewise-
1216 affine trajectory, and the contraction bound $|\Delta\Phi'| \leq \|c\|_\infty \|\Delta v'\|_1$ is Hölder's inequality. (i) An
1217 event with unique optima on both sides is a transversal optimal-vertex switch; its slope jump
1218 satisfies $c^\top \Delta v' = \Delta\Phi' \neq 0$ unless the value function is affine through it, which would force the
1219 optimum to persist — hence objective-invisibility ($c^\top \Delta v' = 0$, $\Delta v' \neq 0$) occurs exactly inside
1220 ≥ 1 -dimensional optimal faces, where the tie-break selects among vertices without moving the
1221 objective; Theorem 2.11(ii) is the sparse-objective special case. The regularity input (continuity
1222 and piecewise affinity of v^*) is measured, not assumed: segment residuals $\leq 8 \times 10^{-14}$ (§4.1,
1223 glucose sweep) and Φ piecewise affine at 4.2×10^{-13} (§4.7). $\square$

1224 B.6 Proof of Proposition 2.12 (collapse)

1225 If f is λ -semiconvex, then $D^2f + \lambda I \cdot \text{Leb} \succeq 0$ as symmetric-matrix measures. For continuous
1226 piecewise-affine f , D^2f is supported on the codimension-one skeleton, hence purely singular with
1227 respect to Lebesgue; testing the inequality on skeleton sets forces the singular part itself PSD,
1228 so $D^2f \succeq 0$. Mollifying, $D^2(f * \rho_\varepsilon) = (D^2f) * \rho_\varepsilon \succeq 0$, and $f * \rho_\varepsilon \rightarrow f$ uniformly, so f is
1229 convex as a uniform limit of convex functions. The converse is trivial. The numerical form: the
1230 smallest violating scale $h_{\max}(\lambda)$ of the OR-plus-cap value function satisfies $h_{\max} = 1/(2\lambda)$ — i.e.
1231 $\lambda h_{\max} = 1/2$, the measured 0.500 identity of the OR-plus-cap value function below — exactly,
1232 i.e. no finite semiconvexity constant exists at any scale. The signed-layer statement is elementary
1233 distribution theory: a continuous PWL function has distributional D^2 supported on its crease
1234 skeleton, a finite symmetric Radon measure with per-crease sign flags. $\square$

1235 B.7 Proof of Proposition 2.13

1236 At a generic codimension-two vertex of a convex PWL f on $\mathbb{R}^2$, exactly three facets meet with
1237 gradients n_1, n_2, n_3 (cyclically ordered). The Monge–Ampère atom is the volume of the normal
1238 fan $\text{conv}\{n_i\}$ (the gradient image sweeps the triangle). Writing $j_i = n_{i+1} - n_i$ for the codimension-
1239 one jumps, twice the fan area is $|\det(j_1, j_2)|$ (the shoelace formula on the triangle), giving atom
1240 $= \frac{1}{2} |\det(j_1, j_2)| = \frac{1}{2} |j_1| |j_2| \sin \angle$; the product $|j_1| |j_2|$ overestimates by $2/\sin \angle \geq 2$ with equality
1241 iff $j_1 \perp j_2$. The simplex-fan generalization to dimension p is the same shoelace argument on a
1242 p -simplex fan. Machine verification: 400 random max-of-affine vertices, determinant = fan area
1243 to 8.9×10^{-16} , the $2/\sin \angle$ law to 2.5×10^{-12} , orthogonal edge case exactly 2. $\square$

1244 B.8 Proof of Theorem 3.1

1245 (i). The interpolant u_h is continuous piecewise affine, so the facet computation of the proof of
1246 Proposition 2.7(i) applies verbatim: D^2u_h is the finite signed measure $\sum_F [\nabla u_h]_F \otimes n_F \mathcal{H}^{p-1}|_F$
1247 over the interior facets of $\mathcal{T}_h$, with the jump constant on each facet (hence the atom $A_F =$
1248 $[\nabla u_h]_F \otimes n_F |F|$ of Definition 2.1, and $[\nabla u_h]_F = [\partial_n u_h]_F n_F$ by continuity of u_h). No atoms sit
1249 on lower-dimensional strata: the mass in $B_\varepsilon(v)$ around a point v of a codimension-two stratum
1250 is at most $\sum_{F \ni v} |[\partial_n u_h]_F| \mathcal{H}^{p-1}(F \cap B_\varepsilon(v)) = O(\varepsilon^{p-1}) \rightarrow 0$, each facet being a $(p-1)$ -polytope
1251 whose intersection with the ball is contained in a $(p-1)$ -ball of radius ε .

1252 (ii). For $\varphi \in C_c^\infty(\Omega)$, $\langle D^2u_h, \varphi \rangle = \int_\Omega u_h D^2\varphi \, \text{dvol}$ by the definition of distributional deriva-
1253 tives; no boundary terms arise because φ is compactly supported. The interpolation bound
1254 $\|u - u_h\|_\infty \leq \frac{1}{2} h^2 |u|_{C^2}$ holds on shape-regular families by a barycentric argument: for $x \in \tau$ with
1255 vertices z_i and barycentric coordinates $\lambda_i(x)$, $u_h(x) - u(x) = \sum_i \lambda_i(x)(u(z_i) - u(x))$, and Taylor's
1256 formula at x gives $u(z_i) - u(x) = \nabla u(x)^\top (z_i - x) + R_i$ with $|R_i| \leq \frac{1}{2} |u|_{C^2} |z_i - x|^2$; the linear terms
1257 sum to $\nabla u(x)^\top (\sum_i \lambda_i(x) z_i - x) = 0$ (the barycenter identity), leaving $\|u_h - u\|_\infty \leq \frac{1}{2} h^2 |u|_{C^2}$.
1258 Hence $\int (u_h - u) D^2\varphi \rightarrow 0$ and $\langle D^2u_h, \varphi \rangle \rightarrow \int u D^2\varphi = \langle \mu, \varphi \rangle$. For the flat-norm statement,
1259 the Kantorovich–Rubinstein norm is the dual norm over the class $\{\phi : \text{Lip}(\phi) \leq 1, \|\phi\|_\infty \leq 1\}$
1260 (Villani, 2009); on the compact Ω this class is uniformly equicontinuous and uniformly bounded,
1261 hence totally bounded in the uniform norm (Arzelà–Ascoli). Weak convergence holds against each

1262 element of a uniform ε -net of the class by the first part of this item, and the uniform mass bound
 1263 of part (iii) controls the remainder of the class, so the supremum over the whole class converges:
 1264 $\|\mu_h - \mu\|_{\text{KR}} \rightarrow 0$, componentwise and hence entrywise. No convergence of total variation is used
 1265 or needed, consistent with part (iii).

1266 (iii). For the upper bound, the gradient of the linear interpolant satisfies, on each simplex τ
 1267 and for every $x \in \tau$,

$$\nabla u_h|_\tau - \nabla u(x) = \sum_i R_i(x) \nabla \lambda_i, \quad |R_i(x)| \leq \frac{1}{2} h^2 |u|_{C^2(\tau)},$$

1268 a decomposition obtained from the same barycentric representation, using $\sum_i \nabla \lambda_i = 0$ and
 1269 $\sum_i (z_i - x) \nabla \lambda_i^\top = I$; shape regularity gives $|\nabla \lambda_i| \leq C/h$, so $\|\nabla u_h - \nabla u\|_{L^\infty(\tau)} \leq Ch|u|_{C^2}$. At
 1270 an interior facet F with midpoint m , $|\nabla u_h|_F| \leq 2Ch|u|_{C^2}$, and $|F| \leq Ch^{p-1}$; quasi-uniformity
 1271 bounds the number of facets by $C|\Omega|h^{-p}$. Summing the entrywise norms of the atoms, $\|\mu_h\|_{\text{TV}} \leq$
 1272 $C|u|_{C^2}|\Omega|$, uniformly in h . For the liminf, the total variation of a matrix-valued measure is the
 1273 sum of the total variations of its entries, each lower semicontinuous under weak convergence
 1274 (Billingsley, 1999), so $\liminf_h \|\mu_h\|_{\text{TV}} \geq \|\mu\|_{\text{TV}}$. That convergence of the total variation itself
 1275 fails in general is the content of Remark 3.2 (exact ratios $3 - 1/n$ and $1.4 - 1/n$; generic smooth
 1276 maps 3.6–4.4, including convex data). In the one-signed regime along a one-dimensional family
 1277 the telescoping identity restores it: for convex $u \in C^2$ on $[a, b]$ with mesh $a = t_0 < \dots < t_N = b$,
 1278 the interpolant u_h is convex, its slope process u'_h is nondecreasing piecewise constant, and

$$\|\mu_h\|_{\text{TV}} = \mu_h([a, b]) = \sum_k \Delta u'_h(t_k) = u'_h(b) - u'_h(a) \rightarrow u'(b) - u'(a) = \mu([a, b]) = \|\mu\|_{\text{TV}},$$

1279 the endpoint slopes being one-sided secant slopes of u ; the concave case is identical up to sign.
 1280 The parametric-LP value function is concave, so its one-dimensional restrictions sit in this regime,
 1281 as does κ^μ along a C^1 trajectory.

1282 (iv). The reconstruction statement is proved in Appendix C (quadratic exactness of the
 1283 per-cell facet sum on structured complexes, a third-order Taylor remainder bound, and the L^1
 1284 summation over cells).

1285 (v). At fixed h , μ_h is a fixed finite facet sum; for the mollified family and a test ψ , $\int \psi d(\mu_h * \varphi_\sigma)$
 1286 $= \int (\psi * \tilde{\varphi}_\sigma) d\mu_h \rightarrow \int \psi d\mu_h$ as $\sigma \rightarrow 0$, because $\psi * \tilde{\varphi}_\sigma \rightarrow \psi$ uniformly and μ_h is a finite
 1287 measure: the limit is the atomic measure. For the window, fix σ and let $h \rightarrow 0$ first:

$$\mu_{h,\sigma}(x) = \langle \mu_h, \varphi_\sigma(x - \cdot) \rangle \rightarrow \langle \mu, \varphi_\sigma(x - \cdot) \rangle = (\mu * \varphi_\sigma)(x),$$

1288 uniformly in x over compact sets, by the weak convergence of part (ii) applied to the family of
 1289 translates $\{\varphi_\sigma(x - \cdot)\}$, which is uniformly bounded and equicontinuous in C^2 , together with the
 1290 uniform mass bound of part (iii). The second limit is classical mollification of the smooth density:
 1291 $\|\mu * \varphi_\sigma - D^2 u\|_{L^1} \leq C\sigma|u|_{C^3}$. Combining the two: at matched resolution, with h/σ small enough
 1292 that the first error is dominated and σ/L_{var} small enough that the second is, the coarse-grained
 1293 family $\mu_{h,\sigma}$ tracks the smooth curvature density, while for $\sigma \ll h$ it resolves the individual atoms
 1294 and for $\sigma \gtrsim L_{\text{var}}$ it averages out the very variation it is supposed to measure: the smooth object
 1295 exists only inside the window $h \ll \sigma \ll L_{\text{var}}$, as measured in §4.6. $\square$

1296 B.9 Proof of Proposition 3.3

1297 The proof combines three facts. (a) *Exact polyhedral identity*. For a polyhedral surface, the
 1298 angle defect at a vertex equals the area of the spherical image of its normal fan: transporting
 1299 the incident facet normals to the unit sphere, the spherical polygon they span has area 2π minus
 1300 the sum of the exterior angles at the vertex, which is the defect (Regge, 1961; Cheeger et al.,
 1301 1984); the identity is machine-verified to 0.0 on the canonical fan (defect, spherical excess, and

fan integral all equal). (b) *Smooth graph identity*. For the graph $S_u = \{(x, u(x))\}$, the Gauss map $n(x) = (-\nabla u(x), 1)/\sqrt{1 + |\nabla u|^2}$ has spherical Jacobian per unit *projected* (parameter) area

$$|\text{Jac } n| = K \frac{dA}{d\text{vol}} = \frac{\det D^2 u}{(1 + |\nabla u|^2)^{3/2}},$$

because the Jacobian with respect to *surface* area is the Gaussian curvature $K = \det D^2 u / (1 + |\nabla u|^2)^2$ (the determinant of the shape operator, the classical Gauss-map identity) and the surface element is $dA = \sqrt{1 + |\nabla u|^2} d\text{vol}$. (c) *Convergence*. The vertex defect measure of the interpolant's graph assigns to each parameter-domain vertex the spherical area of its normal fan; by (a) this is the polyhedral Gauss-map image, and the curvature measures of piecewise-flat approximations converge weakly to the smooth curvature measures (Cheeger et al., 1984). On the graph of u_h over the triangulated parameter domain, the facet normals track the smooth Gauss map at scale h (the gradient interpolation bound $\|\nabla u_h - \nabla u\|_\infty \leq Ch|u|_{C^2}$ of the proof of Theorem 3.1(iii)), so the union of the vertex normal fans over a parameter region R covers the spherical image $n(R)$ up to a vanishing boundary discrepancy. Combining (a)–(c), against a test ψ on the parameter domain,

$$\int \psi dK_h \rightarrow \int \psi(x) \frac{\det D^2 u(x)}{(1 + |\nabla u(x)|^2)^{3/2}} d\text{vol},$$

the Gauss-map area density per projected parameter area. The *intrinsic* Gaussian curvature is the same integrand divided by $\sqrt{1 + |\nabla u|^2}$; the discrepancy factor is exactly $\sqrt{1 + |\nabla u|^2} = \sec(\text{tilt})$, the tilt being the angle between the graph normal and the vertical: the sec law. Recovering intrinsic curvature therefore requires multiplying by the graph area correction, as stated. The 0.0% excess across the 13 probes spanning tilts 0.11–2.83 and three tilt directions is the numerical form of the exact factor. $\square$

B.10 Proof of Proposition 3.5

Smooth case. For a C^2 map the curvature mass is the total variation of $D^2 u$, whose density is the bounded matrix field $D^2 u$: the mass captured by a planar loop of radius ε is at most $\|D^2 u\|_{L^\infty}$ times the loop's area $\pi\varepsilon^2$ — $B(\varepsilon) = O(\varepsilon^2)$, the measured slope 2.000 on the smooth arm of the regime-dial design.

Piecewise-affine case. For a piecewise-affine map the mass measure $D^2 v$ is the facet-layer measure of the proof of Proposition 2.7(i): finitely many facets, each carrying a bounded density against $(p - 1)$ -dimensional facet measure. A loop of radius ε meets any given facet line in a chord of length at most 2ε , so the captured mass is at most density times 2ε per facet crossed: $B(\varepsilon) = O(\varepsilon)$. When the kink complex meets the loop *transversally*, the crossing is genuine, the chord is of exact order ε , and the captured mass is $\Theta(\varepsilon)$: the log-log slope 1.00 of the parametric-FBA arm (76-pair perturbation scan). In general parameter dimension p the same facet-measure computation gives $B(\varepsilon) = O(\varepsilon^{p-1})$; the dichotomy as stated is the planar case, which is where it is measured. $\square$

C Technical proofs

The two long arguments deferred from Appendix B.

C.1 Proof of Lemma 2.10

Well-definedness. For each θ the feasible set $P(\theta) = \{v : Sv = 0, \ell(\theta) \leq v \leq u(\theta)\}$ is a nonempty compact polytope (hypothesis). Stage 1 maximizes a linear objective over $P(\theta)$; its argmax $P_1(\theta)$ is a nonempty compact face. Stage 2 maximizes the secondary objective over $P_1(\theta)$, and so on.

Two selection rules complete the lexicographic stack, and we state both. The *coordinate-pinned completion* minimizes the coordinates $v_1, \dots, v_n$ one at a time, in a fixed order, over the current face: after the coordinate v_j has been fixed to its conditional minimum, the remaining set is the face intersected with the hyperplane $v_j = m_j$; after all n coordinates every component is pinned to a specific value, so the final set is a single point $v^*(\theta)$ — uniqueness is unconditional. The *engine's weighted stage 3* (§7) minimizes $w^\top v$ over the compact stage-2 face with the fixed weight vector w (seed 20240901): for stage-3 objectives of this form the argmin is a unique vertex of the face for every θ outside a Lebesgue-null set of weight vectors (the finitely many hyperplane arrangements $\{w^\top v = w^\top v'\}$, $v \neq v'$ vertices of the face, bound the exceptional set), so the fixed seed realizes a generic draw and the pinned vertex is unique for it; the chamber and projection argument below then applies verbatim, and the coordinate-pinned rule is the representative for which uniqueness holds for *every* w . Existence at every stage follows from compactness, and the lexicographic optimum is therefore well defined under both completions.

Piecewise affinity and continuity. Consider the graph

$$G = \{(v, \theta) : Sv = 0, \ell(\theta) \leq v \leq u(\theta)\} \subset \mathbb{R}^n \times \mathbb{R}^p,$$

a polyhedron, since ℓ and u are affine. The optimal-set graphs are tracked through the stages by induction on the stage index k , with the following claim.

Claim. For each objective stage k there is a finite partition of the parameter box into closed polyhedral chambers $\{K_{k,\alpha}\}$ such that the stage- k value Φ_k is affine on each chamber and the stage- k graph $G_k = \{(v, \theta) \in G_{k-1} : c_k^\top v = \Phi_k(\theta)\}$ intersects $\mathbb{R}^n \times K_{k,\alpha}$ in a polyhedron.

For $k = 1$ this is the structure theorem of multi-parametric linear programming: for an LP whose parameters enter affine right-hand sides, the parameter space admits a finite closed polyhedral chamber cover on which the value is affine and the optimal face varies polyhedrally (Borrelli et al., 2003). Explicitly, over a chamber K with $\Phi_1 = \ell_K$ affine,

$$G_1 \cap (\mathbb{R}^n \times K) = \{(v, \theta) : (v, \theta) \in G, \theta \in K, c_1^\top v = \ell_K(\theta)\},$$

an intersection of polyhedra; the affine constraint $c_1^\top v - \ell_K(\theta) = 0$ is legitimate because $\Phi_1 = \ell_K$ on K , so for $\theta \in K$ membership in the stage-1 face is exactly this equation. For the induction step, write each polyhedron $Q = G_k \cap (\mathbb{R}^n \times K_k)$ in the form $\{(v, \theta) : Av \leq b + B\theta\}$ (every affine constraint in (v, θ) can be so rearranged, with the chamber constraint on θ absorbed into the right-hand side). The stage- $(k+1)$ value $\Phi_{k+1}(\theta) = \max\{c_{k+1}^\top v : Av \leq b + B\theta\}$ is again a multi-parametric LP value function, so it subdivides K_k into finitely many closed sub-chambers on which it is affine and the stage- $(k+1)$ graph is polyhedral, exactly as at $k = 1$ (Borrelli et al., 2003). This proves the claim for the objective stages: each value Φ_k is continuous piecewise affine, and each graph G_k is a finite union of polyhedra, polyhedral over each chamber of a common refinement.

The coordinate-fixing stage keeps polyhedrality by Fourier–Motzkin elimination (Ziegler, 1995). Let $Q = G_k \cap (\mathbb{R}^n \times K)$ be one chamber piece of the stage- k graph and let j be the next coordinate to fix. Consider

$$A_j = \{((v, \theta), (v', \theta')) \in Q \times Q : \theta = \theta', v_j \leq v'_j\},$$

a polyhedron in $\mathbb{R}^{n+p} \times \mathbb{R}^{n+p}$. Its projection onto the first factor is

$$\{(v, \theta) \in Q : v_j \leq v'_j \text{ for every } (v', \theta') \in Q \text{ with } \theta' = \theta\},$$

which is precisely the stage graph after fixing $v_j = m_j(\theta) = \min\{v'_j : (v', \theta) \in Q\}$; projections of polyhedra are polyhedra (Fourier–Motzkin elimination), so the fixed-coordinate graph is polyhedral over the chamber, with compact slices preserved. Iterating over the n coordinates, the final graph G^* is, over each final chamber K_α , a polyhedron that projects injectively onto K_α , the v -slice being the single point $v^*(\theta)$.

Over each final chamber, v^* is the inverse of an injective polyhedral projection. Its continuity is the closed-graph lemma: the graph piece is closed, the slices are singletons, and a function

with closed graph and locally bounded values is continuous. Its piecewise affinity is polyhedral geometry: the face lattice of the graph polyhedron projects to a finite polyhedral decomposition of K_α , and on the relative interior of each projected face the inverse map is affine, an injective affine map between polytopes having an affine inverse on its image (Ziegler, 1995). Chambers overlap on shared faces, and the selections computed on different chambers agree there because $v^*(\theta)$ is unique; gluing the affine pieces along the common closed faces gives a globally defined continuous piecewise-affine v^* . The graph over each GPR cell is a finite union of the chamber pieces (the chambers subdivide the cells, since the LP data is affine over a cell), that is, a finite polyhedral complex, as stated. The measured regularity inputs — path segment residuals $\leq 8 \times 10^{-14}$ (§4.1) and Φ piecewise affine at 4.2×10^{-13} (§4.7) — are the numerical form of the exact polyhedral identities above. $\square$

C.2 Proof of Theorem 3.1(iv)

Structured complexes. Fix a translation-invariant complex with period h : each cell C carries a fixed simplex pattern, and the assignment $F \rightarrow C$ of skeleton facets to cells is translation-equivariant. Write $S_C(u) = \sum_{F \rightarrow C} A_F$ for the per-cell signed facet sum and $\rho_h(C) = |C|^{-1} S_C(u)$ for the reconstructed density. Two properties drive the argument.

(a) *Quadratic exactness.* The map $u \mapsto S_C(u)$ is linear, since interpolation is linear in the data; and S_C kills affine functions, since the interpolant of an affine function is exact and no jumps arise. For the quadratic $q_H(x) = \frac{1}{2}x^\top Hx$, scaling homogeneity of the construction (atoms have the form $[\nabla q]|F| \sim h \cdot h^{p-1}$) gives $S_C(q_H) = c(P)h^p H$ with a dimensionless constant $c(P)$ depending only on the cell pattern. For the structured right-triangle lattice, $c(P)h^p = |C|$: the per-cell atoms $A_{\text{diag}}, A_{\text{vert}}, A_{\text{horiz}}$ of Remark 3.2 sum *exactly* to $h^2 H$ for the interpolant of $u = \frac{1}{2}x^\top Hx$ (machine-verified to 0.0). Hence $S_C(q) = |C| D^2 q$ exactly for every quadratic q : for quadratic data the reconstruction is exact, $\rho_h(C) = D^2 q$.

(b) *Third-order remainder.* Let $u \in C^3$ and q its second-order Taylor polynomial at the cell center x_C . On C , $|u - q| \leq Ch^3|u|_{C^3}$ and $|u - q|_{C^2} \leq Ch|u|_{C^3}$. On each simplex of the pattern, the barycentric gradient bound of the proof of Theorem 3.1(iii) gives $\|\nabla(u - q)_h\|_{L^\infty} \leq \|\nabla(u - q)\|_{L^\infty} + Ch|u - q|_{C^2} \leq Ch^2|u|_{C^3}$, so each facet atom of the remainder is bounded by $Ch^2 \cdot h^{p-1} = Ch^{p+1}$, and each cell carries a fixed number of facets: $|S_C(u - q)| \leq Ch^{p+1}|u|_{C^3}$.

(c) *L^1 summation.* By (a) and (b),

$$\rho_h(C) = |C|^{-1}(S_C(q) + S_C(u - q)) = D^2 u(x_C) + O(h|u|_{C^3}),$$

using $D^2 q = D^2 u(x_C)$ and $|C| \sim h^p$. Summing over cells,

$$\begin{aligned} \|\rho_h - D^2 u\|_{L^1} &= \sum_C \int_C |\rho_h(C) - D^2 u(x)| \, \text{dvol} \\ &\leq \sum_C |C| \left(Ch|u|_{C^3} + \sup_{x \in C} |D^2 u(x) - D^2 u(x_C)| \right) \leq 2Ch|\Omega| |u|_{C^3}, \end{aligned}$$

the second term using $\sup_{x \in C} |D^2 u(x) - D^2 u(x_C)| \leq h|u|_{C^3}$. This is strong L^1 convergence at rate $O(h)$ for $u \in C^3$, and consequently $|\|\rho_h\|_{L^1} - \|D^2 u\|_{L^1}| \leq \|\rho_h - D^2 u\|_{L^1} \rightarrow 0$ with $\|D^2 u\|_{L^1} = \|\mu\|_{\text{TV}}$: the no-mass-loss statement in its correct, cell-averaged form.

Unstructured complexes. The unweighted three-facet patch of an arbitrary cell is not the discrete-duality cell: the correct pairing weights each skeleton facet by its covolume (dual cell) in the normal direction, which is the framework in which the translation-covariance argument above is replaced by the discrete-duality (dual-diamond) identities (Cheeger et al., 1984). The theorem therefore asserts the structured case, proved above, and records the pairing required for the general statement. $\square$

Data, Software, and Code Availability

All figures, tables, and numerical claims in this manuscript are produced by committed scripts from deposited artifacts; the complete bundle (generation scripts, result artifacts, genome-scale models, data manifests with SHA-256 checksums, and the verification scripts that re-derive every manuscript number) is available at [github.com/ MIKEAA2020/metabolic-curvature-measure](https://github.com/MIKEAA2020/metabolic-curvature-measure). Genome-scale models (iJO1366, iML1515) were obtained from the BiGG database; expression compendia from M3D and PRECISE; proteome data from PaxDb and the condition-dependent proteome of Schmidt et al.; all sources are public and cited in the reference list. No new experiments were performed.

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GLM (Z.ai) and DeepSeek AI assisted with development and documentation of analysis code. All AI-assisted content was reviewed, verified, and edited by the author, who takes responsibility for the final content.

Author Contributions

A.A. conceived the study design, developed the methodology and analysis code, performed the data analysis, drafted and revised the manuscript.

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